Data Analyst – Forex Trading Signal AI Agent

Final Project Report

An AI-Accelerated Quantitative FX Trading Signal System: Exploratory Analysis, Technical Signals, Backtesting, Machine Learning, Reinforcement Learning, Risk Management and Business Intelligence (Research Prototype)

Certificate

This is to certify that the project report entitled “Data Analyst – Forex Trading Signal AI Agent” is a bona fide record of work carried out by TIWARI KISHLAYNATH SHAILENDRAKUMAR (Enrollment No. 23193404010045) under my guidance.

Signature: [PLACEHOLDER] Name: [INSERT] Designation: [INSERT] Date: [INSERT] Institutional Seal: [PLACEHOLDER]

Declaration

I declare that this report is my original work, that all results reported are drawn from the project artefacts and screenshots referenced herein, and that no experimental result, dataset or financial performance figure has been fabricated. Where evidence was unavailable, this is stated explicitly.

Signature: [PLACEHOLDER] Date: [INSERT]

Acknowledgement

The author thanks [INSERT GUIDE NAME], the faculty of Monark University, and [INSERT ANY ORGANISATION OR MENTORS] for guidance, feedback and access to tools used in the project. The author also acknowledges the open-source communities behind the analytical software employed. [EDIT AS REQUIRED]

Executive Summary

This project developed an AI-accelerated quantitative foreign-exchange (FX) trading signal prototype. It combines exploratory data analysis (EDA), technical analysis, a weighted composite signal engine, backtesting, machine learning (ML), reinforcement learning (RL), historical case studies, risk management, an R implementation, a Power BI dashboard, DAX Studio validation, Excel validation, a gamification framework and handover documentation.

Critical limitation. The only dataset available contained EUR/USD and approximately 500 rows covering a short period. The project therefore did **not** use five currency pairs or two years of history. All findings are exploratory and cannot be generalised.

The following results are directly visible in the supplied DAX Studio and Power BI screenshots:

- The signal table contains **500** observations: **246 BUY (49.2%)**, **143 SELL (28.6%)** and **111 HOLD (22.2%)**, matching the Python distribution reported in the project brief.
- The Power BI measures return a Sharpe Ratio of **0.9188**, an average return of **5.242×10^{-5}**, Historical VaR (95%) of **0.001402** and Historical VaR (99%) of **0.002024**.
- The ML prediction measures count **91** predictions (**48** correct; TP 1, TN 47, FP 1, FN 42). The derived accuracy of 52.7% is close to chance and the model almost never predicts the positive class.

- The Risk_Fact table contains Parametric, Historical and Monte Carlo VaR, and CVaR, at 95% and 99%; Kelly = 0.07 and Half-Kelly = 0.03 are displayed (two decimals) on the dashboard.
- The screenshots also reveal presentation defects (for example, a donut chart showing 500 for each category and several KPI cards rounding to 0.00) that are documented transparently in Chapter 24.0.

Backtest metrics, ML metrics beyond the confusion counts, RL performance and case-study P&L are **Not available in the provided project evidence** and are not reported. The system must not be interpreted as a profitable or guaranteed trading system.

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Abbreviations

Abbreviation	Meaning
ATR	Average True Range
CVaR / ES	Conditional Value at Risk / Expected Shortfall
DAX	Data Analysis Expressions
DQN	Deep Q-Network
EDA	Exploratory Data Analysis
EMA / SMA	Exponential / Simple Moving Average
FX	Foreign Exchange
LSTM	Long Short-Term Memory network
MACD	Moving Average Convergence Divergence
OHLCV	Open, High, Low, Close, Volume
PBIX	Power BI report file
PPO	Proximal Policy Optimization
RL	Reinforcement Learning
RSI	Relative Strength Index
SHAP	SHapley Additive exPlanations
VaR	Value at Risk

Reading conventions: evidence classification

Every result in this report is classified so that implemented work, validated results, assumptions and limitations remain distinguishable.

Table 0.1 – Evidence classification used in this report

Class	Meaning
E1 – Screenshot-verified	The value or behaviour is visible in an embedded figure (Figures 1–12).
E2 – Author-reported implementation	The work is described in the project brief; no exhibit was supplied.
E3 – Assumption	A modelling or business assumption, not an observed result.
E4 – Not available	Marked “Not available in the provided project evidence”.

1.0 Introduction

1.1 Background

Foreign exchange is the largest financial market by turnover, trading continuously across regional sessions. Price series are noisy, weakly stationary and sensitive to macroeconomic events. Analysts commonly combine technical indicators, statistical risk measures and, more recently, machine-learning and reinforcement-learning methods to support decisions. Business-intelligence tooling then communicates these outputs to non-technical stakeholders.

1.2 Project Overview

The project, titled “Data Analyst – Forex Trading Signal AI Agent”, was executed as a sequence of daily work packages: EDA (Day 2), technical indicators and composite signals (Day 3), backtesting (Day 4), machine learning (Day 5), reinforcement learning (Day 6), case studies (Day 7), risk management (Day 8), R implementation (Day 9), Power BI (Days 10–11), DAX Studio validation (Day 12), gamification and business model (Day 13), end-to-end validation (Day 14) and handover (Day 15). Day 1 is **Not available in the provided project evidence**.

1.3 Nature and Status of the System

The system is an analytical and research prototype. It is not a production trading system, is not connected to a broker, and does not execute orders. It must not be described as guaranteed profitable.

1.4 Report Organisation

Chapters 2.0–8.0 define the problem, scope, requirements, architecture and data. Chapters 9.0–17.0 present the analytical work. Chapters 18.0–24.0 cover Power BI, validation and gamification. Chapters 25.0–32.0 discuss results, limitations, ethics, future work, handover and appendices.

2.0 Problem Statement

Discretionary FX analysis is slow, inconsistent and difficult to audit. Individual indicators frequently disagree; for example, a trend-following indicator may signal BUY while an oscillator signals SELL. Practitioners also tend to report returns without risk context, and analytical results are frequently not reconciled across tools.

The problem addressed is therefore threefold: (i) how to combine several technical signals into a transparent, rule-based composite; (ii) how to compare that composite against ML and RL alternatives under realistic cost and risk considerations; and (iii) how to present and reconcile the results across Python, R, Power BI, DAX Studio and Excel so that figures can be audited.

3.0 Project Objectives

The overarching objective was to develop an AI-accelerated quantitative FX signal system. The specific objectives were:

1. Perform exploratory data analysis and data-quality assessment on OHLCV data.
2. Implement a technical indicator engine and a weighted composite signal.
3. Backtest MA-only and composite strategies with transaction costs, slippage and spread.
4. Build and compare ML models (Random Forest, XGBoost, LSTM) with explainability.
5. Develop an RL trading environment and compare PPO with DQN.
6. Document four historical FX case studies and their implications for risk.
7. Implement risk measures (VaR, CVaR, Kelly sizing).

8. Reproduce key components in R, including a Shiny prototype.
9. Build a Power BI data model, 120 DAX measures and six dashboard pages.
10. Validate measures through DAX Studio and Excel, and reconcile across platforms.
11. Define a gamification framework and an illustrative business model.
12. Produce a final report and handover package.

4.0 Project Scope

Table 4.1 – Scope summary

Area	In scope	Out of scope / not demonstrated
Instruments	EUR/USD (the only pair available)	Multi-pair analysis; cross-pair correlation
Data span	Approximately 500 rows, short period	Two-year history
Analytics	EDA, indicators, composite signal, backtest, ML, RL	Live execution, broker integration
Risk	Single-pair VaR, CVaR, Kelly	Multi-pair correlated Monte Carlo
BI	Power BI model, six pages, DAX Studio checks	Real-time refresh, row-level security
Commercial	Illustrative framework only	Any actual revenue or customer claims

5.0 Business and Analytical Requirements

5.1 Business Requirements

Stakeholders require a traceable signal summary, clear risk indicators, comparisons between strategies, and a dashboard that non-technical users can filter by date, session and signal type.

5.2 Analytical Requirements

The analytics must be reproducible in at least two languages, use chronological validation to avoid look-ahead bias, include transaction costs, and express risk using established statistical measures.

Table 5.1 – Requirements and traceability

ID	Requirement	Addressed in	Evidence
R1	Data-quality assessment	Ch. 9.0	E2
R2	Weighted composite signal	Ch. 11.0	E1 (counts), E2
R3	Cost-aware backtest	Ch. 12.0	E2; metrics E4
R4	ML with explainability	Ch. 13.0	E1 (counts), E2
R5	RL agent with held-out evaluation	Ch. 14.0	E2; metrics E4
R6	Risk measures	Ch. 16.0	E1
R7	R reproduction	Ch. 17.0	E2
R8	Power BI dashboard	Ch. 18.0	E1
R9	DAX Studio reconciliation	Ch. 20.0	E1
R10	Handover package	Ch. 30.0	E2

6.0 System Architecture

6.1 Objective

To describe the logical layers and data flow of the prototype.

6.2 Methodology and Implementation

The system follows a layered pipeline. Raw OHLCV data are cleaned and enriched in the EDA layer (output: forex_eda_processed.csv). The indicator and signal layer produces forex_signals.csv. Backtesting, ML and RL layers consume these files. Risk metrics are computed separately. Outputs are loaded into Power BI fact and dimension tables and validated in DAX Studio and Excel.

Table 6.1 – System architecture layers

Layer	Function	Principal artefacts
Ingestion	Load OHLCV data	Raw dataset (source: not specified in the evidence)
Processing / EDA	Cleaning, returns, rolling volatility	forex_eda_processed.csv
Signal engine	Indicators, component signals, composite score	forex_signals.csv
Strategy evaluation	Backtesting, ML, RL	Python and R code
Risk	VaR, CVaR, Kelly	Risk_Fact
Presentation	Power BI, R Shiny prototype	PBIX file, Shiny app
Validation	DAX Studio, Excel, cross-language checks	Validation queries and workbook

[INSERT FIGURE P-1: End-to-End System Architecture Diagram]

6.3 Limitations

The architecture is batch-oriented; no streaming ingestion, scheduler or deployment infrastructure is evidenced.

7.0 Data Architecture

7.1 Objective

To define the analytical data model used in Power BI.

7.2 Implementation

Eight tables were created. In the DAX Studio screenshots the physical model objects carry the prefix “PowerBI_” (for example PowerBI_Signal_Dim and PowerBI_Risk_Fact); this report uses the logical names.

Table 7.1 – Power BI data model tables

Logical table	Type	Role
Date_Dim	Dimension	Calendar and time attributes
Price_Fact	Fact	OHLCV, spread and derived price fields
Technical_Dim	Dimension	Indicator values (SMA, EMA, RSI, MACD, ATR, etc.)
Signal_Dim	Dimension	Component signals and Composite_Signal
ML_Fact	Fact	ML predictions and outcomes
RL_Fact	Fact	RL observations and actions
Risk_Fact	Fact	Metric / Value rows for VaR, CVaR, Kelly
Case_Fact	Fact	Case-study attributes

Relationships connect Date_Dim to the fact and dimension tables. The specific cardinalities and filter directions are **Not available in the provided project evidence**.

[INSERT FIGURE P-2: Power BI Model View – Relationships]

8.0 Dataset Description

8.1 Objective

To characterise the dataset honestly, including its limitations.

8.2 Description

The dataset contains OHLCV observations for **EUR/USD** only, with fields DateTime, Open, High, Low, Close, Volume and Spread. The project brief states approximately 500 rows. The dashboards corroborate 500 observations at hourly resolution beginning on 01 January 2023 (Figures 4–7); the weekday totals in Figure 7 (72 observations for most weekdays and 68 for Saturday) imply roughly three weeks of data. The data provider and licence are **Not available in the provided project evidence**.

Table 8.1 – Dataset characteristics

Characteristic	Value	Evidence
Instrument	EUR/USD only	E2

Characteristic	Value	Evidence
Observations	500	E1 (Figures 1, 10)
Frequency	Hourly (inferred from hour-of-day visuals)	E1 (Figures 6, 7)
Start date	01 January 2023 (first slicer date)	E1
Average Close, all rows	1.088007	E1 (Figure 11)
Average Spread, all rows	1.4056 (unit not specified)	E1 (Figure 11)
Observations by session	Asia 168; London 105; London–New York Overlap 105; New York 82; Off Session 40	E1 (Figure 7)
Weekend observations	Present (Saturday 68, Sunday 72 in Figure 7)	E1

8.3 Limitations

500 hourly rows span a very short, non-diversified period. The sample cannot represent multiple volatility regimes, central-bank cycles or liquidity conditions. No statistical conclusion should be extrapolated beyond this sample. Cross-pair analysis is impossible. The presence of weekend rows is unusual for spot FX and is treated as a data-quality observation (Chapter 9.0).

9.0 Data Quality and EDA

9.1 Objective

To establish whether the OHLCV data are structurally valid and statistically characterised before signal generation, and to document defects that affect interpretation.

9.2 Methodology

The OHLCV file was loaded and the fields DateTime, Open, High, Low, Close, Volume and Spread were inspected. Returns were computed, descriptive moments were calculated, and a sequence of integrity checks was applied. A processed dataset, **forex_eda_processed.csv**, was exported for downstream stages.

9.3 Technical Explanation and Formulas

The one-period simple return and the logarithmic return are defined as follows. The report does not state which variant the project code used; the variant is therefore **Not available in the provided project evidence**.

$$r_t = \frac{C_t - C_{t-1}}{C_{t-1}} \qquad r_t^{\log} = \ln\left(\frac{C_t}{C_{t-1}}\right)$$

Descriptive statistics (mean, standard deviation, skewness and excess kurtosis) were calculated on the return series:

$$\mu = \frac{1}{n}\sum_{t=1}^{n} r_t, \quad \sigma = \sqrt{\frac{1}{n-1}\sum_{t=1}^{n}(r_t-\mu)^2}, \quad S = \frac{1}{n}\sum_{t=1}^{n}\left(\frac{r_t-\mu}{\sigma}\right)^3, \quad K = \frac{1}{n}\sum_{t=1}^{n}\left(\frac{r_t-\mu}{\sigma}\right)^4 - 3$$

Rolling volatility over a window of w observations is the sample standard deviation of the most recent w returns. The window length is **Not available in the provided project evidence**.

$$\sigma_t^{(w)} = \sqrt{\frac{1}{w-1}\sum_{i=0}^{w-1}\left(r_{t-i} - \bar{r}_t^{(w)}\right)^2}$$

Excess kurtosis greater than zero indicates fatter tails than a normal distribution, and non-zero skewness indicates asymmetry. These properties matter because the parametric VaR in Chapter 16.0 assumes normality.

9.4 Implementation and Results

Table 9.1 – Data-quality checks performed

Check	Performed	Outcome	Evidence
Missing values	Yes	Not available in the provided project evidence	E2 / E4
Duplicate rows	Yes	Not available in the provided project evidence	E2 / E4
Invalid OHLC relationships (for example High < Low)	Yes	Not available in the provided project evidence	E2 / E4

Check	Performed	Outcome	Evidence
Weekend observations	Yes	Present: Saturday 68 and Sunday 72 observations (Figure 7)	E1
Return statistics (mean, SD, skewness, kurtosis)	Yes	Numeric values not available in the provided project evidence	E2 / E4
Rolling volatility	Yes	Series computed; values not available in the provided project evidence	E2 / E4
Rolling cross-pair correlation	Attempted	Not meaningfully computable: only EUR/USD available	E2

The only numerically visible summary statistics are those exposed through Power BI and DAX Studio: Average Close 1.088007, Average Spread 1.4056 (Figure 11), Average Daily Return $5.24208806577353 \times 10^{-5}$ (Figure 10), and an annualised volatility displayed as 0.01 after rounding (Figure 5).

[INSERT FIGURE P-3: Return Distribution Histogram, Return Series and Rolling Volatility (from forex_eda_processed.csv)]

9.5 Interpretation

The existence of weekend observations is notable, because spot FX is normally closed from Friday evening to Sunday evening. The observations may reflect the data vendor's convention, but this cannot be confirmed from the supplied evidence. Because the signal engine consumed the same rows, weekend bars are included in all downstream counts (the weekday totals in Figure 7 sum to 500).

9.6 Limitations

The sample is small and short. Moment estimates (particularly skewness and kurtosis) from approximately 500 hourly returns are statistically unstable. Cross-pair correlation, a central requirement for portfolio-level risk, could not be evaluated.

10.0 Technical Indicator Engine

10.1 Objective

To compute a standard set of trend, momentum, volatility and cloud indicators from EUR/USD prices for use in the signal engine and as ML features.

10.2 Methodology

Indicators were implemented first in Python and reproduced in R (Chapter 17.0). Parameters stated in the project brief are SMA 20 and 50, EMA 20 and 50, RSI 14 and ATR 14. Parameters for MACD, Bollinger Bands and Ichimoku are **Not available in the provided project evidence**; the conventional values quoted below are reference definitions, not confirmed project settings.

Table 10.1 – Technical indicators and parameters

Indicator	Category	Stated parameters	Conventional reference parameters (E3)
SMA 20, SMA 50	Trend	20, 50	–
EMA 20, EMA 50	Trend	20, 50	–
RSI 14	Momentum	14	Wilder smoothing
MACD, Signal, Histogram	Momentum / trend	Not stated	12, 26, 9
Bollinger Bands	Volatility	Not stated	20 periods, 2 standard deviations
Ichimoku components	Trend / cloud	Not stated	9, 26, 52
ATR 14	Volatility	14	–

10.3 Technical Explanation and Formulas

$$SMA_N(t) = \frac{1}{N}\sum_{i=0}^{N-1} C_{t-i}$$

$$EMA_N(t) = \alpha C_t + (1-\alpha)\,EMA_N(t-1), \qquad \alpha = \frac{2}{N+1}$$

$$RSI = 100 - \frac{100}{1+RS}, \qquad RS = \frac{\text{average gain over 14 periods}}{\text{average loss over 14 periods}}$$

$$MACD = EMA_{fast}(C) - EMA_{slow}(C), \quad Signal = EMA_{9}(MACD), \quad Histogram = MACD - Signal$$

$$BB_{mid} = SMA_{N}(C), \quad BB_{up/low} = BB_{mid} \pm k\,\sigma_N(C)$$

$$TR_t = \max\{H_t - L_t,\ |H_t - C_{t-1}|,\ |L_t - C_{t-1}|\}, \qquad ATR_{14}(t) = \text{smoothed average of } TR \text{ over 14 periods}$$

The Ichimoku components comprise the Tenkan-sen and Kijun-sen (midpoints of the highest high and lowest low over short and medium windows), Senkou Span A and B (the projected cloud boundaries), and the Chikou span (lagged close).

10.4 Results

Indicator values are held in Technical_Dim. The only values visible in the supplied exhibits are the averages on the Pair Analysis page for the date selected in the slicer (04 January 2023): Average SMA 20, SMA 50, EMA 20 and EMA 50 each display 1.07; Average RSI displays 56.60; Average MACD and Average ATR display 0.00 (Figure 6). MACD and ATR are small-magnitude quantities for EUR/USD, so a two-decimal display format is likely to be rounding them to zero; the underlying values should be inspected in DAX Studio.

[INSERT FIGURE P-4: EUR/USD Close with SMA/EMA, Bollinger Bands and Ichimoku Overlays]

10.5 Interpretation and Limitations

An average RSI of 56.60 sits between the conventional oversold (30) and overbought (70) thresholds, which indicates neutral-to-mildly-bullish momentum for the filtered date only. Moving averages with 50-period windows require 50 observations to initialise, so the first rows carry undefined values; the treatment of these warm-up rows is **Not available in the provided project evidence**. With only about 500 rows, the warm-up consumes a material fraction of the sample.

11.0 Composite Signal Engine

11.1 Objective

To convert seven indicator-level signals into one transparent, weighted decision score with BUY, SELL and HOLD outcomes.

11.2 Methodology

Each component (SMA, EMA, RSI, MACD, Bollinger Band, Ichimoku and ATR) emits a discrete signal s_i in {-1, 0, +1}. The component signals are combined with fixed weights into a score, and the score is mapped to a class using symmetric thresholds. The output was saved as **forex_signals.csv**.

Table 11.1 – Composite signal weights and thresholds

Component	Weight
SMA Signal	0.15
EMA Signal	0.15
RSI Signal	0.15
MACD Signal	0.20
Bollinger Band Signal	0.10
Ichimoku Signal	0.15
ATR Signal	0.10
Sum	**1.00**

Condition	Class
Score ⩾ 0.30	BUY
Score ⩽ −0.30	SELL
Otherwise	HOLD

11.3 Technical Explanation and Formulas

```
Score_t = \sum_{i=1}^{7} w_i\, s_{i,t}, \qquad \sum_i w_i = 1, \qquad Score_t \in [-1, +1]
```

Because the weights sum to one, the score is bounded in [-1, +1]. MACD carries the largest weight (0.20), reflecting its role as a combined trend-momentum indicator. The ATR signal and Bollinger Band signal carry the lowest weights (0.10 each). The exact rule used to assign -1, 0 or +1 for each component is **Not available in the provided project evidence**. A conventional illustration (E3, not confirmed) is: SMA 20 above SMA 50 gives +1; RSI below 30 gives +1 and above 70 gives -1; MACD above its signal line gives +1.

The 0.30 threshold implies that a BUY requires a net weighted agreement of at least 30%. For instance, SMA, EMA and MACD all at +1 would yield 0.50 before other components are added.

11.4 Implementation and Results

Table 11.2 – Composite signal distribution (500 observations)

Class	Count	Share of total	Evidence
BUY	246	49.2%	E1 (Figures 1, 10, 11, 12) and Python
SELL	143	28.6%	E1 and Python
HOLD	111	22.2%	E1 and Python
Total	500	100%	E1

The percentages are computed from the displayed counts. The DAX measures reproduce the Python counts exactly (246 / 143 / 111), as shown in Figure 1. The code in the screenshot is truncated after the SELL measure, so the HOLD definition is not visible; the HOLD result of 111 is visible in the output.

```
EVALUATE
ROW(
    "Total Signals",
        COUNTROWS(PowerBI_Signal_Dim),

    "BUY Signals",
        CALCULATE(
            COUNTROWS(PowerBI_Signal_Dim),
            PowerBI_Signal_Dim[Composite_Signal] = "BUY"
        ),

    "SELL Signals",
        CALCULATE(
            COUNTROWS(PowerBI_Signal_Dim),
            PowerBI_Signal_Dim[Composite_Signal] = "SELL"
```

Log **Results** History

Total Signals	BUY Signals	SELL Signals	HOLD Signals
500	246	143	111

Figure 1 – DAX Studio query and result: Total Signals (COUNTROWS of PowerBI_Signal_Dim) and CALCULATE-based BUY and SELL counts filtered on Composite_Signal; output 500 / 246 / 143 / 111. *Source: project DAX Studio session (E1).*

11.5 Interpretation

The distribution is BUY-heavy (49.2%) compared with SELL (28.6%). Figure 4 shows the EUR/USD close rising across the displayed period, which is consistent with trend-following components contributing positive signals more often. This is a descriptive observation about one short sample, not evidence of predictive skill. Correlation between components (for example SMA and EMA, which both derive from the same close) also means the effective diversification of the seven components is lower than the weight structure suggests.

11.6 Limitations

Weights and thresholds were fixed by design, not estimated; no sensitivity analysis is available in the supplied evidence. Component rules are undocumented here. The class balance is specific to this sample.

12.0 Backtesting Framework

12.1 Objective

To compare an MA-only strategy against the composite-signal strategy under cost-aware, chronologically ordered evaluation.

12.2 Methodology

Two strategies were implemented: (i) an MA-only strategy and (ii) the composite-signal strategy. Costs comprised transaction cost, slippage and spread cost. A walk-forward moving-average optimisation was carried out using a 70/30 chronological train/validation structure, so that parameters were selected on the earlier 70% of observations and assessed on the later 30%, avoiding look-ahead leakage.

Table 12.1 – Backtesting strategies and cost components

Element	Description	Numeric setting
MA-only strategy	Position from moving-average relationship	Not available in the provided project evidence
Composite strategy	Position from BUY / SELL / HOLD class	Not available in the provided project evidence
Transaction cost	Per-trade fee	Not available in the provided project evidence
Slippage	Execution deviation	Not available in the provided project evidence
Spread cost	Bid–ask cost, using the Spread field	Not available in the provided project evidence
Validation split	70% train, 30% validation, chronological	70/30 (E2)

12.3 Technical Explanation and Formulas

A signal observed at the close of bar t is applied to bar t+1 to prevent look-ahead bias. With position p_t in {-1, 0, +1}:

$$R^{strat}_{t+1} = p_t\, r_{t+1} - c\,|p_t - p_{t-1}|, \qquad c = c_{txn} + c_{slip} + c_{spread}$$

Performance metrics are defined as follows.

$$TotalReturn = \prod_{t}(1+R^{strat}_t) - 1, \qquad WinRate = \frac{N_{winning\ trades}}{N_{trades}}$$

$$Sharpe = \frac{\bar{R}}{\sigma_R}\sqrt{A}, \qquad ProfitFactor = \frac{\text{gross profit}}{\text{gross loss}}, \qquad MDD = \min_t\left(\frac{E_t}{\max_{s\le t}E_s} - 1\right)$$

Here A is the annualisation factor (periods per year) and E_t is the equity curve. For hourly data A depends on the trading-hours convention, which is **Not available in the provided project evidence**.

12.4 Results

Table 12.2 – Backtesting metrics and evidence status

Metric	MA-only	Composite
Total Return	Not available in the provided project evidence	Not available in the provided project evidence
Win Rate	Not available in the provided project evidence	Not available in the provided project evidence
Sharpe Ratio	Not available in the provided project evidence	Not available in the provided project evidence
Profit Factor	Not available in the provided project evidence	Not available in the provided project evidence
Maximum Drawdown	Not available in the provided project evidence	Not available in the provided project evidence

The Sharpe Ratio of 0.9188 in the Power BI and DAX Studio exhibits (Figures 5, 10 and 12) is a dashboard measure. Its input series (raw price returns versus strategy returns) is not documented in the supplied evidence, and it must **not** be read as the backtest Sharpe Ratio of either strategy.

[INSERT FIGURE P-5: Equity Curve and Drawdown – MA-only versus Composite Strategy]

12.5 Interpretation and Limitations

In a sample of roughly three weeks, the number of trades is small, so win rate and profit factor would have wide confidence intervals. Walk-forward optimisation on 350 training rows and 150 validation rows (the 70/30 split of 500 observations) is a weak test of parameter stability. Any backtest outcome is hypothetical, assumes fills at modelled prices, and does not guarantee future performance.

13.0 Machine Learning Models

13.1 Objective

To test whether engineered features derived from EUR/USD price, volume and time information can support a supervised signal, and to compare tree-based and sequence models under time-aware validation.

13.2 Methodology

More than 30 features were engineered. Models were trained with chronological validation to avoid leakage, then compared, and feature influence was examined with SHAP. The prediction target definition (for example next-bar direction), class encoding, train/test boundaries and hyper-parameters are **Not available in the provided project evidence**.

Table 13.1 – Feature groups

Group	Description
Return lags	Past returns at multiple lags
Rolling volatility	Rolling standard deviation of returns
Candle geometry	Body size, wicks and range derived from Open, High, Low, Close
Volume ratios	Volume relative to a rolling reference
Bollinger Band features	Position within bands and band width
Time / session features	Hour, weekday and trading-session indicators

Table 13.2 – Models implemented

Model	Validation approach	Notes
Random Forest	TimeSeriesSplit	Bagged decision trees; also reproduced in R with caret
XGBoost	Early-stopping based workflow	Gradient-boosted trees; also reproduced in R
LSTM	Sequence input	Recurrent neural network
Model comparison	Common evaluation	Per-model metrics not available in the provided project evidence

13.3 Technical Explanation and Formulas

TimeSeriesSplit creates expanding training windows, each followed by a later validation block, so a model is never validated on data that precede its training data. Random Forest splits nodes by reducing Gini impurity:

```
G = 1 - \sum_{k} p_k^2
```

XGBoost minimises a regularised objective, where l is a loss, f_k are trees, and Omega penalises tree complexity:

```
\mathcal{L} = \sum_i l(y_i, \hat{y}_i) + \sum_k \Omega(f_k), \qquad \Omega(f)
= \gamma T + \tfrac{1}{2}\lambda \lVert w \rVert^2
```

An LSTM cell updates its state through forget, input and output gates:

$$f_t = \sigma(W_f[h_{t-1},x_t]+b_f),\quad i_t = \sigma(W_i[h_{t-1},x_t]+b_i),\quad c_t = f_t \odot c_{t-1} + i_t \odot \tanh(W_c[h_{t-1},x_t]+b_c)$$

SHAP attributes a prediction to features through Shapley values, where F is the feature set and S a subset:

$$\phi_j = \sum_{S \subseteq F\setminus\{j\}} \frac{|S|!\,(|F|-|S|-1)!}{|F|!}\left[f(S\cup\{j\}) - f(S)\right]$$

Classification measures used below are defined from the confusion counts:

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN},\quad Precision = \frac{TP}{TP+FP},\quad Recall = \frac{TP}{TP+FN},\quad F1 = \frac{2\,Precision\,Recall}{Precision+Recall}$$

13.4 Results

The only ML outputs in the supplied evidence are the Power BI prediction counts retrieved in DAX Studio (Figure 2). The producing model is not identified in the exhibit. The derived measures in Table 13.3 are computed by the report author from those counts and are not separately reported project results.

```
60      "Average Spread", [Average Spread]
61  )
62
63  EVALUATE
64  ROW(
65      "Total ML Predictions", [Total ML Predictions],
66      "Correct Predictions", [Correct Predictions],
67      "False Positives", [False Positives],
68      "False Negatives", [False Negatives],
69      "True Positives", [True Positives],
70      "True Negatives", [True Negatives]
71  )
72
73  EVALUATE
74  ROW(
```

Log **Results** History

Total ML Predictions	Correct Predictions	False Positives	False Negatives	True Positives	True Negatives
91	48	1	42	1	47

Figure 2 – DAX Studio: ML prediction measures. Total ML Predictions 91; Correct Predictions 48; False Positives 1; False Negatives 42; True Positives 1; True Negatives 47. *Source: project DAX Studio session (E1).*

Table 13.3 – ML confusion counts and derived measures

Quantity	Value	Basis
Total predictions	91	Displayed
True Positives / True Negatives	1 / 47	Displayed
False Positives / False Negatives	1 / 42	Displayed
Correct predictions	48	Displayed (equals TP + TN)
Accuracy	48 / 91 = 52.7%	Derived
Precision	1 / 2 = 50.0%	Derived
Recall (sensitivity)	1 / 43 = 2.3%	Derived
Specificity	47 / 48 = 97.9%	Derived
F1 score	approximately 4.4%	Derived
Actual positives / actual negatives	43 / 48	Derived
Majority-class (all-negative) accuracy	48 / 91 = 52.7%	Derived

The counts are internally consistent: 1 + 47 + 1 + 42 = 91 and 1 + 47 = 48.

13.5 Interpretation

The model predicted the positive class only twice in 91 cases (TP + FP = 2). Its accuracy of 52.7% is identical to that of a trivial classifier that always predicts the negative class, and its recall is 2.3%. The Strategy Comparison page (Figure 9) is consistent with this: the ML Signal Distribution shows a single bar at -1. The evidence therefore indicates that, for the model and test window shown, **no useful discriminative skill was demonstrated**. The cause cannot be determined from the supplied material; plausible contributors include a small sample, weak predictive content in lagged features, class imbalance and threshold choice, but these remain hypotheses. Whether the positive class corresponds to BUY or to another target is **Not available in the provided project evidence**.

[INSERT FIGURE P-6: SHAP Summary Plot and Model Comparison Chart]

13.6 Limitations

ML metrics for Random Forest, XGBoost and LSTM individually, SHAP rankings and hyper-parameters are not available. Approximately 500 rows is very small for an LSTM and for 30+ features; over-fitting risk is high and the 91-prediction test set gives imprecise estimates. Financial returns have low signal-to-noise, so near-chance accuracy is a plausible and honest outcome that should be reported rather than hidden.

14.0 Reinforcement Learning Agent

14.1 Objective

To investigate whether an agent can learn a position-taking policy directly from market observations, accounting for trading costs.

14.2 Methodology

A custom Gymnasium environment, **ForexTradingEnv**, was developed. The agent chooses among Long, Short and Flat-style actions based on an observation window of recent market features. Transaction and spread costs enter the reward calculation, and the reward is based on trading performance. PPO and DQN agents were trained and compared, and evaluation used a held-out segment of the data.

Table 14.1 – RL environment specification

Element	Specification	Evidence
Environment	Custom Gymnasium ForexTradingEnv	E2
Actions	Long / Short / Flat-style	E2
Observation	Window of recent features (window length not available in the provided project evidence)	E2
Costs	Transaction / spread cost considered	E2
Reward	Trading-performance based (exact formula not available in the provided project evidence)	E2
Algorithms	PPO; DQN for comparison	E2

Element	Specification	Evidence
Evaluation	Held-out evaluation	E2
Observations recorded in RL_Fact	70	E1 (Figure 12)

14.3 Technical Explanation and Formulas

The task is modelled as a Markov decision process with state s_t (observation window), action a_t and reward r_t. A representative reward is the position return net of cost:

```
r_t = p_t \cdot \Delta C_{t+1}/C_t - c\,|p_t - p_{t-1}|
```

The exact project reward is not stated; the expression above is a generic template. PPO maximises a clipped surrogate objective, where rho_t is the probability ratio between new and old policies and A_t the advantage estimate:

```
L^{CLIP}(\theta) = \mathbb{E}_t\left[\min\left(\rho_t A_t,\ \mathrm{clip}
(\rho_t, 1-\epsilon, 1+\epsilon)A_t\right)\right]
```

DQN learns an action-value function by minimising the temporal-difference error:

```
L(\theta) = \mathbb{E}\left[\left(r + \gamma \max_{a'} Q(s',a';\theta^-) -
Q(s,a;\theta)\right)^2\right]
```

14.4 Results

Quantitative RL performance (cumulative reward, return, drawdown, DQN versus PPO comparison) is **Not available in the provided project evidence**. The only RL outputs visible are those on the Strategy Comparison page (Figure 9). The PPO Signal column totals **53** across the displayed rows, and RL_Fact holds **70** observations. Reading the PPO Signal Distribution chart, the bars correspond to approximately 8 observations at -1, one at 0 and 61 at +1 (visually estimated; these values are consistent with the net total of 53 and 70 observations: 61 - 8 = 53; 61 + 8 + 1 = 70). The PPO policy was therefore long in roughly 87% of the displayed observations.

[INSERT FIGURE P-7: PPO and DQN Training Curves and Held-out Evaluation]

14.5 Interpretation

A predominantly long policy is unsurprising in a short sample where EUR/USD drifted upward (Figure 4) and is not evidence of skill; a constant-long policy would also look favourable under that drift. Moreover, the project's evaluation methodology did not fully mark-to-market open positions, and final open positions were not always force-closed. Reported or implied RL profits are therefore **unreliable and must not be interpreted as realised profitability**. A correct evaluation would value open positions at the final close, force-close at the end of the episode and deduct the corresponding exit cost.

14.6 Limitations

The held-out set is small, results depend on random seeds (not documented), PPO and DQN are sample-hungry, and the single-pair, single-regime data restrict generalisation.

15.0 Forex Market Case Studies

15.1 Objective

To place the system in the context of four historical FX regimes and to extract risk-management lessons.

15.2 Methodology and Evidence Policy

The market descriptions below summarise widely documented public events and are given at an approximate level; figures should be verified against the primary sources listed in Chapter 31.0. The "signal behaviour" discussion is analytical reasoning about how a trend-weighted composite would be expected to behave (E3) and is **not** the result of running the model on case-study data. Model portfolio P&L for all four cases is **Not available in the provided project evidence**, and none is reported. The Case_Fact table in Power BI holds case attributes; its contents were not supplied.

Table 15.1 – Case study overview

Case	Date	Regime type	Model P&L
USD/JPY Carry Unwind	July–August 2024	Rapid funding-currency reversal	Not available in the provided project evidence

Case	Date	Regime type	Model P&L
GBP Flash Crash	7 October 2016	Liquidity-driven intraday gap	Not available in the provided project evidence
USD/INR Range Trade	2023–24	Managed, low-volatility range	Not available in the provided project evidence
EUR/CHF Floor Removal	15 January 2015	Policy-regime break	Not available in the provided project evidence

15.3 Case 1 – USD/JPY Carry Unwind (July/August 2024)

Market event. After the Bank of Japan raised its policy rate at the end of July 2024 and weaker US labour data followed, positions funded in yen were reduced rapidly, with a sharp equity-market sell-off in early August.

Market movement. USD/JPY fell by roughly 10% or more from its early-July peak above 160 to levels near 142 in early August (approximate).

Signal behaviour (expected). Trend components (SMA, EMA, MACD, Ichimoku) lag, so BUY signals would be expected to persist into the early part of the reversal, followed by abrupt SELL signals after the damage was done. ATR and Bollinger Bands would register the volatility expansion.

Risk implications. Crowded carry positions create gap and correlation risk across yen crosses, which a single-pair model cannot capture.

Model adaptation. Add volatility-scaled position sizing, a rate-differential regime filter and multi-pair exposure limits.

Lessons learned. Carry returns resemble short-volatility payoffs; stop-loss discipline and cross-pair correlation monitoring are essential.

15.4 Case 2 – GBP Flash Crash (7 October 2016)

Market event. During thin Asian-session liquidity, sterling experienced a very rapid fall and partial recovery, amid Brexit-related headlines and automated selling; the episode was later reviewed by an international central-bank committee.

Market movement. GBP/USD fell by several percent within minutes and recovered most of the move shortly afterwards (approximate).

Signal behaviour (expected). Hourly indicators would barely register an intraday spike that reverses within the bar; a bar-close system might show an extreme wick (captured by candle-geometry features) but little change in the composite score.

Risk implications. Stop orders can execute at poor prices in gaps; spread and slippage assumptions of ordinary conditions understate stress costs.

Model adaptation. Add liquidity-aware cost models, session filters (the Session Heatmap already separates Asia, London, New York, overlap and off-session) and circuit-breaker rules.

Lessons learned. Tail events originate in market microstructure as well as news; historical VaR from calm samples is not a bound on such losses.

15.5 Case 3 – USD/INR Range Trade (2023–24)

Market event. The rupee traded in a comparatively narrow band, with central-bank management of volatility through spot and forward-market operations.

Market movement. Low realised volatility and gradual drift in the low-to-mid 80s per US dollar (approximate).

Signal behaviour (expected). Trend-following components would generate frequent whipsaw signals in a range; mean-reversion components (RSI, Bollinger Bands) would be better suited. Composite scores would hover near the HOLD band.

Risk implications. Low volatility invites leverage, while policy shifts can end the range abruptly.

Model adaptation. Add a range/trend regime classifier and reduce trend-weight in low-ATR regimes.

Lessons learned. A fixed weight scheme is regime-dependent, and managed currencies violate the random-walk assumptions behind parametric VaR.

15.6 Case 4 – EUR/CHF Floor Removal (15 January 2015)

Market event. The Swiss National Bank abruptly discontinued the 1.20 minimum exchange rate that it had maintained since 2011, and cut its deposit rate further into negative territory.

Market movement. EUR/CHF collapsed by roughly 15% to 30% within minutes, intraday, before partially retracing (approximate); several retail FX brokers suffered severe losses.

Signal behaviour (expected). No indicator-based system would anticipate a discontinuity; positions held long EUR/CHF would have been exposed to an unhedgeable gap, and indicators would update only after the event.

Risk implications. A pegged or capped currency has a hidden, asymmetric tail risk that standard deviation and VaR estimated during the peg period understate; negative balances and leverage amplify losses.

Model adaptation. Add stress scenarios, peg and floor detection, leverage caps, and exclusion rules for administratively fixed rates.

Lessons learned. Historical and parametric VaR fail under regime breaks; scenario analysis and conservative position sizing (such as fractional Kelly) are necessary complements.

15.7 Limitations

The case discussions are qualitative. Without data for these periods in the dataset (which holds only a short EUR/USD sample), no replay or P&L was possible, and no claim is made that the model would have avoided or profited from any of the four events.

16.0 Risk Management Framework

16.1 Objective

To quantify potential loss and position-size exposure using parametric, historical and simulation-based measures, and to document the stored risk metrics.

16.2 Methodology

Parametric VaR (95% and 99%), Historical VaR (95% and 99%), Monte Carlo VaR, CVaR (Expected Shortfall), the Kelly Criterion and Half-Kelly sizing were implemented. Losses are expressed as positive fractions of position value per observation period. The Monte Carlo configuration (number of simulations, return distribution, calibration window, random seed) is **Not available in the provided project evidence**.

Important limitation. The available data are single-pair EUR/USD. A correlated multi-pair Monte Carlo simulation, which requires a covariance matrix across currencies, could not be fully demonstrated. All risk figures describe a single-instrument exposure.

16.3 Technical Explanation and Formulas

With return mean mu, standard deviation sigma and standard-normal quantile z_alpha (1.645 at 95%, 2.326 at 99%):

```
VaR^{param}_{\alpha} = z_{\alpha}\,\sigma - \mu
```

Historical VaR is the empirical loss quantile, and CVaR is the mean loss beyond it:

```
VaR^{hist}_{\alpha} = -Q_{1-\alpha}(r), \qquad CVaR_{\alpha} = -
\,\mathbb{E}\left[r \mid r \le -VaR_{\alpha}\right]
```

Monte Carlo VaR applies the same quantile to simulated returns r^(m), for m = 1..M, drawn from a fitted model:

```
VaR^{MC}_{\alpha} = -Q_{1-\alpha}\left(\{r^{(m)}\}_{m=1}^{M}\right)
```

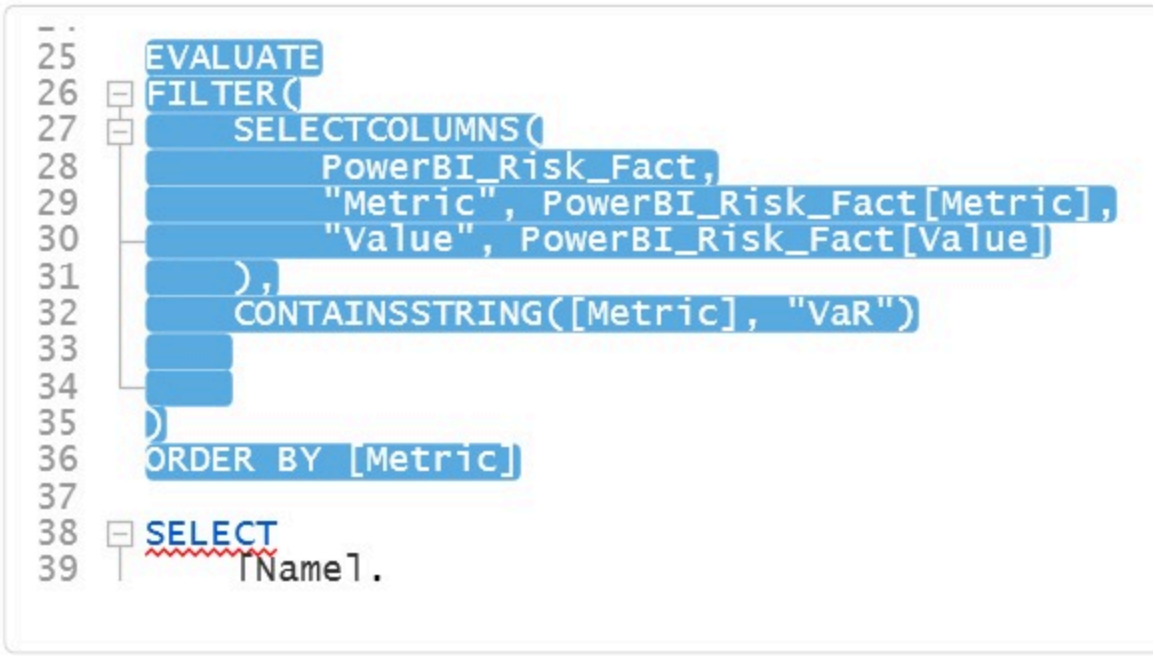
The Kelly fraction maximises expected logarithmic growth. For a strategy with win probability p, loss probability q = 1 - p and payoff ratio b, and the half-Kelly rule scales it by one half to reduce estimation risk:

```
f^{*} = \frac{b\,p - q}{b}, \qquad f^{half} = \tfrac{1}{2} f^{*}
```

The inputs used for Kelly in the project (p, b, or a continuous-time variant) are **Not available in the provided project evidence**.

16.4 Results

Figure 3 shows the Risk_Fact table (Metric and Value columns) filtered to metric names containing “VaR”. Values are shown to six significant digits in Table 16.1.



```
25  EVALUATE
26  FILTER(
27      SELECTCOLUMNS(
28          PowerBI_Risk_Fact,
29          "Metric", PowerBI_Risk_Fact[Metric],
30          "Value", PowerBI_Risk_Fact[Value]
31      ),
32      CONTAINSSTRING([Metric], "VaR")
33
34
35  )
36  ORDER BY [Metric]
37
38  SELECT
39      [Name1].
```

Log **Results** History

Metric	Value
CVaR 95%	0.0017980118467663
CVaR 99%	0.002430652760064
Historical VaR 95%	0.0014022457876813
Historical VaR 99%	0.0020255255397716
Monte Carlo VaR 95%	0.0014478526597632
Monte Carlo VaR 99%	0.0020513539267222
Parametric VaR 95%	0.0014387826311256
Parametric VaR 99%	0.0020566167033302

Figure 3 – DAX Studio: PowerBI_Risk_Fact rows containing "VaR" (CVaR, Historical, Monte Carlo and Parametric VaR at 95% and 99%). *Source: project DAX Studio session (E1).*

Table 16.1 – Risk metrics retrieved from Risk_Fact

Metric	95%	99%
Parametric VaR	0.00143879	0.00205662

Metric	95%	99%
Historical VaR	0.00140225	0.00202553
Monte Carlo VaR	0.00144785	0.00205135
CVaR	0.00179801	0.00243065

Kelly Fraction and Half-Kelly Fraction are displayed as 0.07 and 0.03 on the dashboard (Figure 8, two-decimal display); the Monte Carlo & VaR chart axis places them at approximately 0.069 and 0.034 (visually estimated). Units (loss per observation versus per day) are not labelled in the exhibits.

Because the Historical VaR values in Figure 3 differ slightly from the Historical VaR measures retrieved in Figures 10 and 12, a comparison is given in Table 16.2.

Table 16.2 – Historical VaR: DAX measure versus Risk_Fact table

Confidence	DAX measure (Figures 10, 12)	Risk_Fact value (Figure 3)	Absolute difference	Relative difference
95%	0.00140174897	0.00140224579	approx. 4.97×10^{-7}	approx. 0.035%
99%	0.00202426982	0.00202552554	approx. 1.26×10^{-6}	approx. 0.062%

Differences are computed by the report author from the displayed values.

16.5 Interpretation

The metrics are internally coherent. At each confidence level CVaR exceeds VaR (CVaR/VaR is about 1.28 at 95% and 1.20 at 99% on the historical basis), and 99% VaR exceeds 95% VaR for every method, as expected. The three VaR methods agree within roughly 3% at 95% and 2% at 99%, which suggests that the return distribution in this short sample is not strongly non-normal in the tails examined; however, the kurtosis and skewness values needed to confirm this are not available.

An author-derived consistency check is informative. If the dashboard Sharpe Ratio (0.9188) equals mean/standard deviation multiplied by the square root of 252, then, with the average return of 5.242×10^{-5}, the implied per-observation standard deviation is about 0.000906. Parametric VaR computed from that value would be about 0.00144 at 95% and

0.00205 at 99%, within about 0.1% of the stored values, and an annualised volatility of about 0.0144 would display as 0.01 (Figure 5). This is **consistent with**, but does not prove, a square-root-of-252 annualisation applied to per-observation returns. Since the exhibits indicate hourly observations (Figures 6 and 7) while the measure is labelled “Average Daily Return”, the periodicity and annualisation basis require verification before the Sharpe Ratio or annualised volatility are quoted externally.

The Historical VaR of 0.14% at 95% means that, in this sample, the loss on one observation period exceeded about 0.14% in roughly 5% of cases. It is not a worst-case bound.

A Kelly fraction of about 0.07 (and half-Kelly of about 0.03) is derived from in-sample statistics of a very short period and is highly sensitive to estimation error. It should not be treated as a recommended position size.

16.6 Limitations

Single pair; roughly three weeks of data; no stress testing; VaR is not sub-additive and fails under regime breaks (Chapter 15.6); Monte Carlo settings undocumented; the unit and periodicity of the stored metrics are unlabelled. The small differences in Table 16.2 are most plausibly due to a percentile-interpolation convention or a different input series between the DAX measure and the table, but the cause is **not confirmed** by the supplied evidence.

17.0 R Implementation

17.1 Objective

To reproduce the principal Python analytics independently in R, as a cross-language verification and as a basis for an interactive prototype.

17.2 Methodology and Implementation

The R workflow used tidyverse packages for data processing; calculated the technical indicators; reproduced the composite signal; trained a Random Forest through caret and an XGBoost model; ran a Monte Carlo simulation; computed Historical VaR and CVaR; and produced ggplot2 visualisations. An R Shiny dashboard prototype was developed. Python-versus-R signal validation was performed.

Table 17.1 – Python–R component mapping

Component	Python	R	Cross-check evidence
Data processing	Python scripts	tidyverse	Numeric comparison not available in the provided project evidence
Indicators	Python	R implementations	Numeric comparison not available in the provided project evidence
Composite signal	forex_signals.csv (246 / 143 / 111)	Reproduced in R	R counts not available in the provided project evidence; validation reported as performed
Random Forest	TimeSeriesSplit workflow	caret	Metrics not available in the provided project evidence
XGBoost	Early-stopping workflow	xgboost in R	Metrics not available in the provided project evidence
Monte Carlo and VaR / CVaR	Python	R	Numeric comparison not available in the provided project evidence
Visualisation	Python charts	ggplot2	–
Dashboard	Power BI	R Shiny prototype	See below

17.3 Shiny Dashboard Prototype

The prototype contains a signal summary, a BUY / SELL / HOLD selector, an EUR/USD close-price chart and a signal-activity chart. It serves as a lightweight demonstration of the signal data; it does not replicate the six-page Power BI report and is not deployed to a hosting service in the supplied evidence.

[INSERT FIGURE P-8: R Shiny Dashboard Prototype Screenshot]

17.4 Technical Explanation

In caret, resampling for time-ordered data should use rolling-origin or time-slice schemes rather than random k-fold cross-validation; the resampling scheme used in R is **Not available in the provided project evidence**. Differences between R and Python results may legitimately arise from window-initialisation conventions, smoothing recursions for RSI and ATR, floating-point differences, and percentile definitions (R supports nine quantile types).

17.5 Results, Interpretation and Limitations

The project brief reports that the Python and R signals were validated against each other, but the agreement rate or any discrepancy list is not part of the supplied evidence. Exact equality is therefore not claimed. The language-level reproduction raises confidence that the signal logic is specified unambiguously, but it cannot validate the economic merit of the signals.

18.0 Power BI Architecture

18.1 Objective

To present the analytical outputs in an interactive dashboard that business users can filter by date, month and trading session, and that exposes signals, risk, price behaviour, session activity and strategy comparison.

18.2 Methodology and Implementation

The processed Python outputs were loaded into eight tables (Chapter 7.0), related through Date_Dim, and exposed through 120 DAX measures (Chapter 19.0). Six report pages were built. Visual-to-table mappings stated in this chapter are inferred from field names visible in the exhibits. Common design elements are a page title, KPI cards, date tiles or radio slicers, a month slicer, and an EUR/USD banner. Refresh mechanism, data-source connection type and publishing workspace are **Not available in the provided project evidence**.

Table 18.1 – Dashboard pages

#	Page	Principal purpose	Evidence
1	Signal Overview	Signal counts, composite score trend and price context	Figure 4
2	Risk Management	VaR, volatility, Sharpe Ratio and drawdown indicators	Figure 5
3	Pair Analysis	EUR/USD price, spread, volume and indicator averages	Figure 6
4	Session Heatmap	Activity by trading session and weekday	Figure 7
5	Monte Carlo & VaR	Comparison of VaR, CVaR and Kelly metrics	Figure 8
6	Strategy Comparison	Technical, ML and PPO signals side by side	Figure 9

18.3 Page 1 – Signal Overview

Purpose. Give an at-a-glance view of the composite signal outcome and its evolution over time.

Data used. Signal_Dim (Composite_Signal and the composite score), Price_Fact (close price), Date_Dim.

Visuals. KPI cards for Total Signals (500), SELL Signals (143) and HOLD Signals (111); a bar visual for BUY Signals; a donut chart of signal classes; a line chart of the composite score across January 2023; a line chart of EUR/USD close; date tile slicers; a month card.

KPIs. Total 500; BUY 246 (confirmed by DAX Studio and by Figure 7); SELL 143; HOLD 111.

Interpretation. The composite score oscillates between roughly -0.65 and +0.75 (visually estimated) and repeatedly crosses the ±0.30 thresholds, producing regimes of sustained BUY or SELL. The score does not approach ±1.0, indicating that all seven components rarely agree. The close-price line rises over the displayed period, aligning with the BUY-dominant distribution. **Defect observed:** the donut chart labels each class as “500 (33.33%)”, which contradicts the 246 / 143 / 111 split and indicates that the chart’s value field does not respect the class breakdown (likely a total-count measure used instead of a count by Composite_Signal; not confirmed). The BUY KPI is shown as a bar rather than a card, and the price-axis labels are partially clipped.

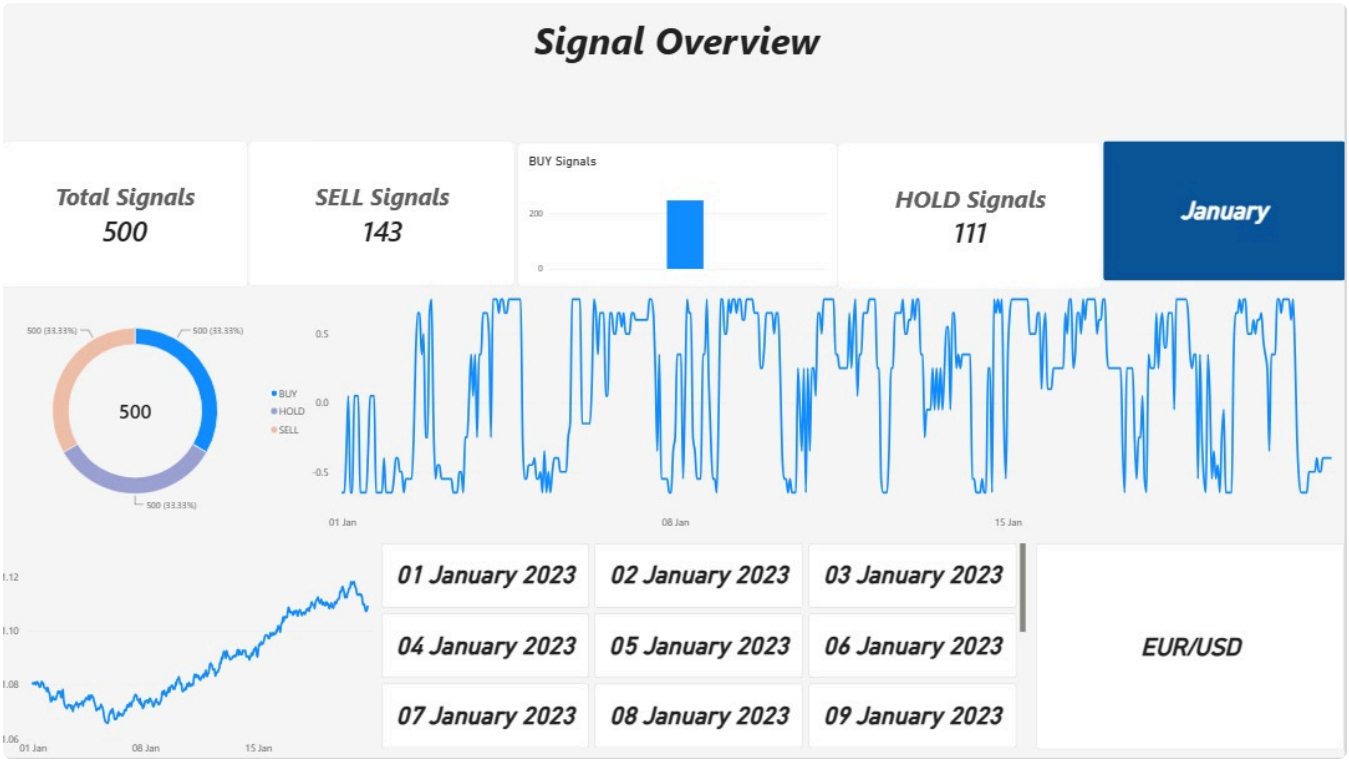


Figure 4 – Power BI Signal Overview page: KPI cards, composite score line chart, EUR/USD close chart and date tiles. *Source: project Power BI report (E1).*

18.4 Page 2 – Risk Management

Purpose. Summarise single-pair market risk and volatility indicators.

Data used. Risk_Fact and Price_Fact (daily range, returns), Date_Dim.

Visuals. Six KPI cards (Parametric VaR 95%, Parametric VaR 99%, Historical VaR 95%, Historical VaR 99%, Return Volatility, Annualized Volatility); a clustered column chart of the four VaR values; a line chart of Average Daily Range by DateTime; a metric table; date and month slicers.

KPIs. Annualized Volatility 0.01; Sharpe Ratio 0.92; Maximum Close Drawdown -0.01; the VaR, Return Volatility, Average Daily Range, Average Gain and Average Loss cards display 0.00.

Interpretation. The column chart shows the 99% VaR bars (about 0.0020–0.0021) above the 95% bars (about 0.0014), matching Table 16.1. **Defect observed:** the five KPI cards display 0.00 although the underlying values are of the order of 0.001–0.002, so the two-decimal number format hides the information the page exists to show. A percentage format with two or three decimals is recommended. The Average Daily Range series fluctuates around roughly 0.001 with occasional peaks near 0.0027 (visual estimate).



Figure 5 – Power BI Risk Management page: VaR KPI cards, VaR comparison chart, Average Daily Range line chart and risk metric table. *Source: project Power BI report (E1).*

18.5 Page 3 – Pair Analysis

Purpose. Describe EUR/USD price levels, spread, trading volume and indicator averages for a selected date.

Data used. Price_Fact, Technical_Dim, Date_Dim.

Visuals. KPI cards (Average Close, Maximum Close, Minimum Close, Average Spread); a line chart of Average Close by DateTime; a dot plot of Average Close, Low, Open and High; a column chart and a line chart of Total Volume by DateTime; a table of average SMA 20, SMA 50, EMA 20, EMA 50, RSI, MACD and ATR; a date slicer; the EUR/USD banner.

KPIs. With 04 January 2023 selected: Average Close 1.07, Maximum Close 1.08, Minimum Close 1.07, Average Spread 1.35, Average RSI 56.60.

Interpretation. The intraday close line declines from a peak near 1.075 around mid-day to about 1.0705 late in the day. Volume is irregular, with a pronounced peak of roughly 8,000 units around 08:00 (units not labelled). The Average Spread of 1.35 for the selected date differs from the all-row value of 1.4056 (Figure 11); this is consistent with the date filter

and does not indicate an error. The two Total Volume charts convey the same information and one is redundant. The Average Close, Low, Open and High dots occupy a very narrow range (about 1.0736–1.0747), so the visual offers little analytical value in its current form.



Figure 6 – Power BI Pair Analysis page for 04 January 2023: price KPIs, intraday close, volume charts and indicator-average table. *Source: project Power BI report (E1).*

18.6 Page 4 – Session Heatmap

Purpose. Show how observations are distributed across trading sessions and weekdays.

Data used. Signal_Dim and Price_Fact joined to a trading-session attribute and Date_Dim.

Visuals. A session-by-weekday matrix with conditional shading; KPI cards (Total 500, BUY 246, SELL 143, HOLD 111); a line chart of Signal Activity by Hour; a column chart of Average of Close by Trading Session; date tiles; month and pair banners.

KPIs. Session totals: Asia 168; London 105; London–New York Overlap 105; New York 82; Off Session 40 (sum 500). Weekday totals: 72 for each weekday except Saturday, which shows 68.

Interpretation. The matrix counts observations per session and weekday; it does not split them by BUY, SELL or HOLD, so the term “signal activity” refers to the number of signal rows. The Asia session shows 24 observations per weekday, London and the overlap 15 each, New York 12 and Off Session 6 (Saturday: 10 and 4 for New York and Off Session).

Assuming three occurrences of each weekday over three weeks, this implies session windows of about 8, 5, 5, 4 and 2 hours per day (an author inference; the session definitions are **Not available in the provided project evidence**). The hourly line chart shows 21 observations for hours 0–19 and 20 for hours 20–23, which sums to 500 and indicates that the series ends at 19:00 on its last day. Saturday and Sunday contain observations, which should be reviewed (Chapter 9.5). The Average Close by session bars are uninformative because the axis starts at zero and the values are about 1.09. **Presentation defects:** the matrix title reads “Seasson wise Signal Activity Heatmap”.

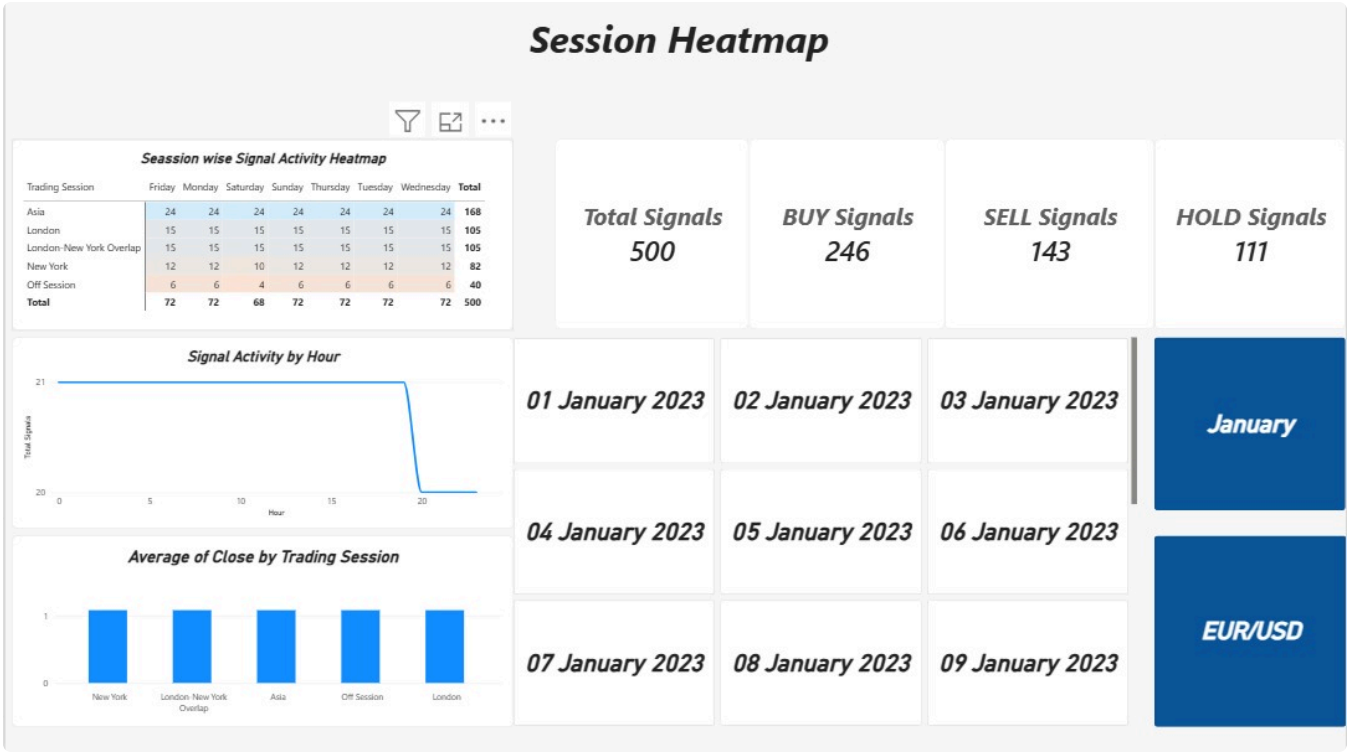


Figure 7 – Power BI Session Heatmap page: signal activity by trading session and weekday, hourly activity and KPI cards. *Source: project Power BI report (E1).*

18.7 Page 5 – Monte Carlo & VaR

Purpose. Compare all stored risk metrics on one page.

Data used. Risk_Fact (Metric and Value columns).

Visuals. A metric table with Sum of Value; KPI cards for Monte Carlo VaR 95% and 99%, CVaR 95% and 99%, Half-Kelly Fraction and Kelly Fraction; a ranked bar chart titled “Risk Metric Comparison by Metric”.

KPIs. Kelly Fraction 0.07; Half-Kelly Fraction 0.03; Monte Carlo VaR and CVaR cards display 0.00.

Interpretation. The ranked bars follow the order Kelly, Half-Kelly, CVaR 99%, Parametric VaR 99%, Monte Carlo VaR 99%, Historical VaR 99%, CVaR 95%, Monte Carlo VaR 95%, Parametric VaR 95%, Historical VaR 95%. This ordering agrees with the values retrieved in DAX Studio (Figure 3), which supports the correctness of the page's data binding.
Defects observed: (i) the KPI cards round VaR and CVaR to 0.00; (ii) the table's Total of 0.12 sums unrelated metrics (loss quantiles and position-size fractions) and has no meaning; (iii) plotting Kelly fractions on the same axis as VaR compresses the VaR bars into near-equal heights, hiding the differences; (iv) the title uses the spelling "Comparision".

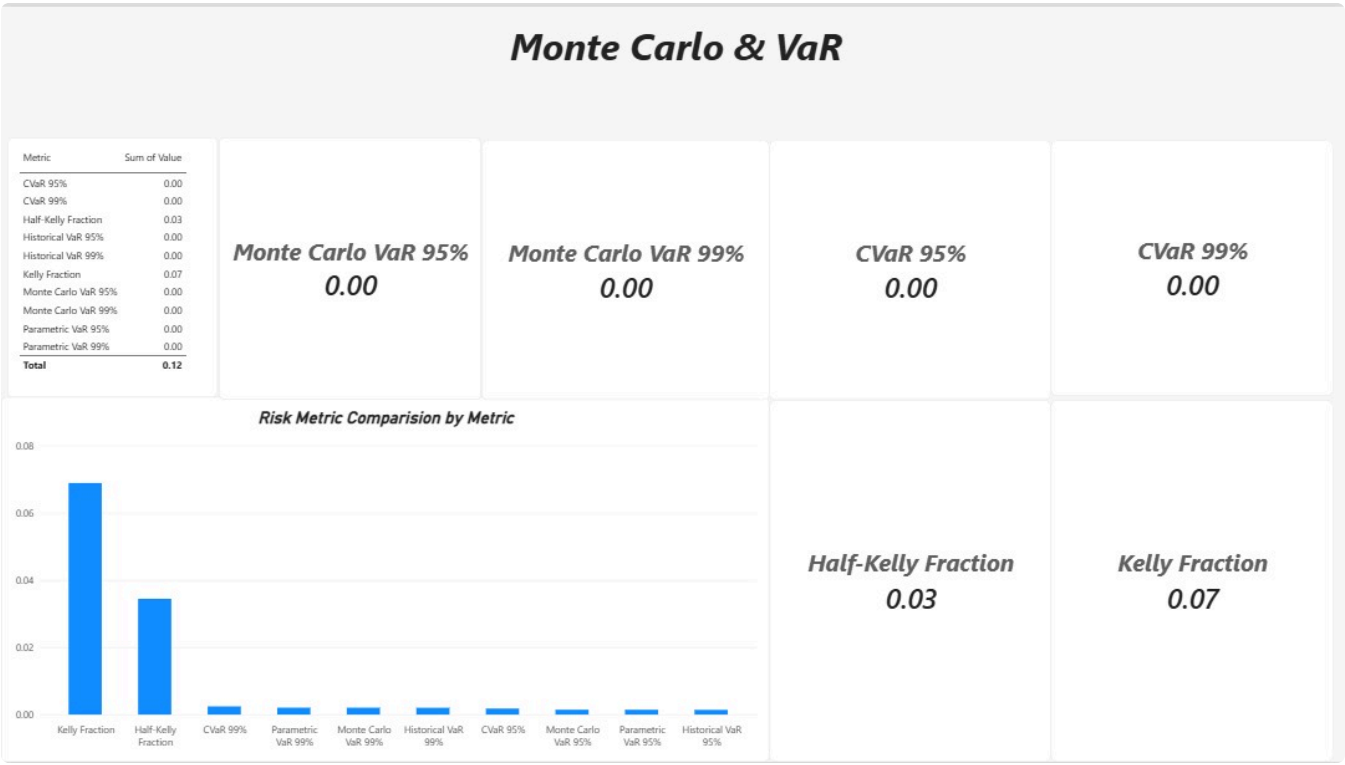


Figure 8 – Power BI Monte Carlo & VaR page: metric table, VaR and CVaR cards, Kelly and Half-Kelly cards and ranked risk-metric chart. *Source: project Power BI report (E1).*

18.8 Page 6 – Strategy Comparison

Purpose. Compare the technical, ML and PPO signals for the same timestamps.

Data used. Signal_Dim, ML_Fact, RL_Fact, Price_Fact (next-period return), Date_Dim.

Visuals. A detail table (DateTime, PPO Signal, ML Signal, Technical Signal, Next Return) with a totals row; three distribution charts (PPO, ML, Technical); a date slicer.

KPIs. Totals row: PPO Signal 53; ML Signal -70; Technical Signal 5; Next Return 0.00.

Interpretation. The totals are net sums of signals (+1, 0, -1). The charts reconcile with them: Technical shows approximately 34 at +1, 7 at 0 and 29 at -1 (net 5); ML shows only -1 (about 70, net -70); PPO is as described in Chapter 14.4. All three charts therefore cover about 70 rows, equal to the RL observation count (Figure 12). In every visible table row the ML signal is -1, matching the near-zero positive-class recall in Chapter 13. The page demonstrates agreement and disagreement between methods, but does **not** demonstrate performance: Next Return displays 0.00 at the selected precision, and no measure aggregates next-period return by signal. **Presentation defects:** the title reads "Strategy Comparision", and the DateTime column uses a different format from the date tiles.

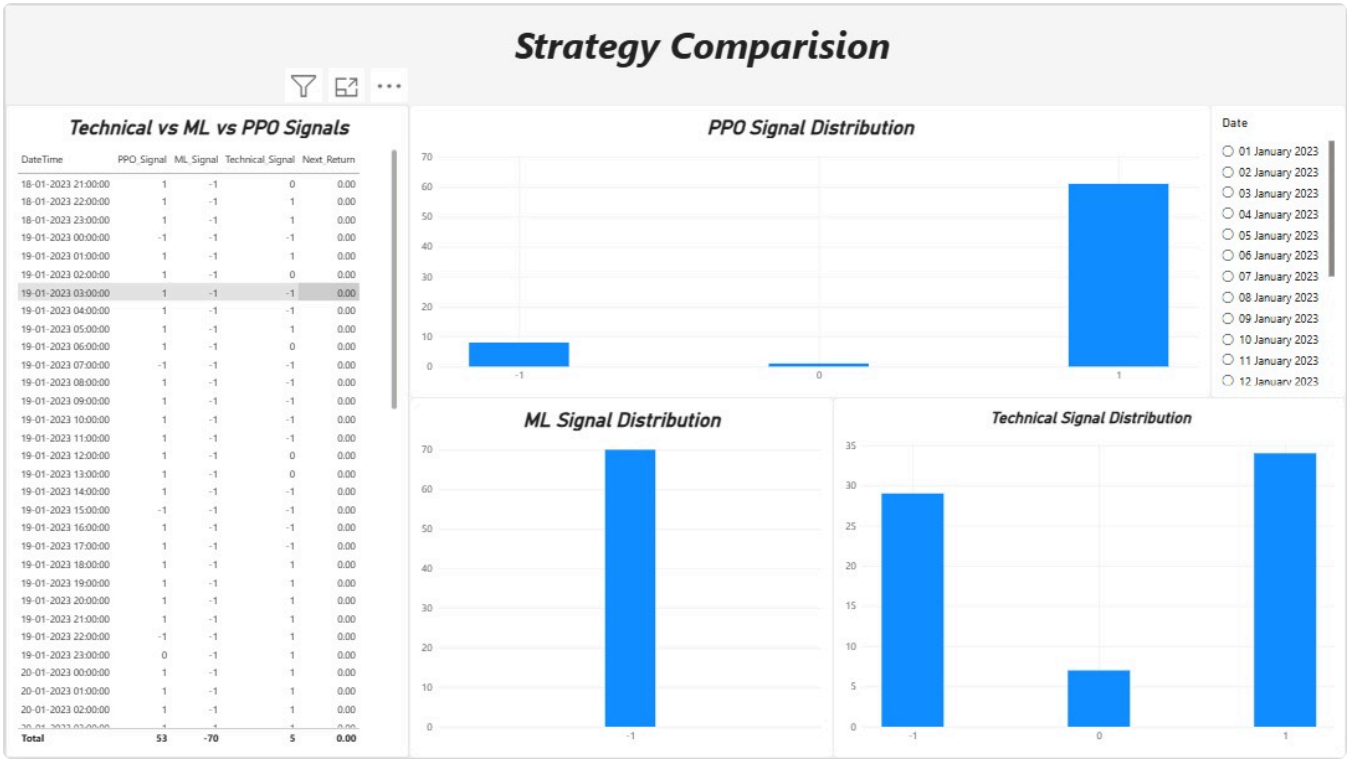


Figure 9 – Power BI Strategy Comparison page: Technical versus ML versus PPO signals and their distributions. *Source: project Power BI report (E1).*

18.9 Limitations

The pages describe a small, single-pair sample. Several KPI formats hide values, one chart (the donut) is incorrect, and some visuals add little information. Refresh, security and sharing configuration are not evidenced.

19.0 DAX Measures

19.1 Objective

To centralise business logic in reusable DAX measures so that every dashboard page and validation query draws on the same definitions.

19.2 Methodology and Implementation

The project reports **120 DAX measures** (author-reported, E2). The count is not visible in the supplied screenshots; Figure 10 shows a query against the model's measure catalogue (SELECT [Name], [TableID] FROM $SYSTEM.TMSCHEMA_MEASURES ORDER BY [Name]) but its result is not shown. Table 19.1 groups the measure names that are visible in the exhibits. It is a partial inventory and must not be read as the complete list of 120.

Table 19.1 – DAX measure families (names visible in the exhibits)

Family	Measures visible in the exhibits
Signal counts	Total Signals, BUY Signals, SELL Signals, HOLD Signals
Price and volume	Average Close, Maximum Close, Minimum Close, Average Open, Average High, Average Low, Average Spread, Total Volume
Technical indicators	Average SMA 20, Average SMA 50, Average EMA 20, Average EMA 50, Average RSI, Average MACD, Average ATR
Risk and performance	Parametric VaR 95% / 99%, Historical VaR 95% / 99%, Monte Carlo VaR 95% / 99%, CVaR 95% / 99%, Kelly Fraction, Half-Kelly Fraction, Return Volatility, Annualized Volatility, Sharpe Ratio, Average Daily Return, Average Daily Range, Maximum Close Drawdown, Average Gain, Average Loss
Machine learning	Total ML Predictions, Correct Predictions, True Positives, True Negatives, False Positives, False Negatives
Reinforcement learning	Total RL Observations

Family	Measures visible in the exhibits
Strategy comparison	PPO Signal, ML Signal, Technical Signal, Next Return (fields or measures)

19.3 Technical Explanation

The signal-count measures follow the standard pattern of counting rows in a context-filtered table. The inline form visible in Figure 1 is:

```
EVALUATE
ROW(
    "Total Signals", COUNTROWS(PowerBI_Signal_Dim),
    "BUY Signals",
        CALCULATE(COUNTROWS(PowerBI_Signal_Dim),
PowerBI_Signal_Dim[Composite_Signal] = "BUY"),
    "SELL Signals",
        CALCULATE(COUNTROWS(PowerBI_Signal_Dim),
PowerBI_Signal_Dim[Composite_Signal] = "SELL")
)
```

CALCULATE changes the filter context, so a visual that slices by Date_Dim automatically recomputes the counts for the selected dates. The query above recomputes the counts directly from the table, which provides an independent cross-check of the stored measures. The definitions of the VaR, Sharpe, ML and RL measures are not visible in the exhibits and are **Not available in the provided project evidence**.

19.4 Results and Interpretation

Every measure retrieved by name in DAX Studio returned a value (Chapter 20.0), which indicates that the measures resolve against the model. Measure outputs reconcile with the table-level recomputation for the signal counts (500 / 246 / 143 / 111). The Risk Management and Monte Carlo pages exposed presentation-format problems rather than calculation problems (Chapter 18.0).

19.5 Limitations and Recommendations

Measure documentation (name, definition, format string, dependencies) is not supplied; a measure dictionary should accompany the PBIX file (Chapter 30.0). Recommended, but not evidenced as implemented: measures for ML accuracy, precision and recall; next-period return by signal class; and percent-formatted VaR measures.

20.0 DAX Studio Validation

20.1 Objective

To validate and reconcile the Power BI measures by querying the model directly, independently of report visuals and their formatting.

20.2 Methodology

DAX Studio was connected to the Power BI model. Queries of the form EVALUATE ROW(...) retrieved several measures at once; an EVALUATE FILTER(SELECTCOLUMNS(...)) query retrieved rows from the risk table; a catalogue query listed the measures. Results were then compared with dashboard visuals and Python counts. This stage is a **validation and reconciliation** exercise, not an independent recomputation of the analytics.

Table 20.1 – DAX Studio validation queries and returned values

Figure	Query content	Returned values
1	COUNTROWS and CALCULATE on PowerBI_Signal_Dim	500 / 246 / 143 / 111
2	ML prediction measures	91 predictions; 48 correct; FP 1; FN 42; TP 1; TN 47
3	PowerBI_Risk_Fact filtered on “VaR”	Eight VaR / CVaR values (Table 16.1)
10	Total Signals, Average Daily Return, Sharpe Ratio, Historical VaR 95% / 99%	500; $5.24208806577353 \times 10^{-5}$; 0.918820176905784; 0.00140174896611285; 0.0020242698225955
11	Signal counts, Average Close, Average Spread	500 / 246 / 143 / 111; 1.088007; 1.4056
12	Signals, ML predictions, RL observations, Historical VaR, Sharpe Ratio	500 / 246 / 143 / 111; 91; 70; 0.00140174896611285; 0.0020242698225955; 0.918820176905784

```
39      [Name],
40      [TableID]
41  FROM $SYSTEM.TMSCHEMA_MEASURES
42  ORDER BY [Name]
43
44  EVALUATE
45  ROW(
46      "Total Signals", [Total Signals],
47      "Average Daily Return", [Average Daily Return],
48      "Sharpe Ratio", [Sharpe Ratio],
49      "Historical VaR 95%", [Historical VaR 95%],
50      "Historical VaR 99%", [Historical VaR 99%]
51  )
52
53  EVALUATE
```

og **Results** History

Total Signals	Average Daily Return	Sharpe Ratio	Historical VaR 95%	Historical VaR 99%
500	5.24208806577353E-05	0.918820176905784	0.00140174896611285	0.0020242698225955

Figure 10 – DAX Studio: measure catalogue query and a ROW query returning Total Signals 500, Average Daily Return 5.24208806577353E-05, Sharpe Ratio 0.918820176905784, Historical VaR 95% 0.00140174896611285 and Historical VaR 99% 0.0020242698225955. *Source: project DAX Studio session (E1).*

```
49          "Historical VaR 95%", [Historical VaR 95%],
50          "Historical VaR 99%", [Historical VaR 99%]
51      )
52
53      EVALUATE
54      ROW(
55          "Total Signals", [Total Signals],
56          "BUY Signals", [BUY Signals],
57          "SELL Signals", [SELL Signals],
58          "HOLD Signals", [HOLD Signals],
59          "Average Close", [Average Close],
60          "Average Spread", [Average Spread]
61      )
62
63      EVALUATE
```

Log **Results** History

Total Signals	BUY Signals	SELL Signals	HOLD Signals	Average Close	Average Spread
500	246	143	111	1.088007	1.4056

Figure 11 – DAX Studio: Total Signals 500, BUY 246, SELL 143, HOLD 111, Average Close 1.088007 and Average Spread 1.4056. *Source: project DAX Studio session (E1).*

```
81  EVALUATE
82  ROW(
83      "Total Signals", [Total Signals],
84      "BUY Signals", [BUY Signals],
85      "SELL Signals", [SELL Signals],
86      "HOLD Signals", [HOLD Signals],
87      "Total ML Predictions", [Total ML Predictions],
88      "Total RL Observations", [Total RL Observations],
89      "Historical VaR 95%", [Historical VaR 95%],
90      "Historical VaR 99%", [Historical VaR 99%],
91      "Sharpe Ratio", [Sharpe Ratio]
92  )
```

Log **Results** History

Total Signals	BUY Signals	SELL Signals	HOLD Signals	Total ML Predictions	Total RL Observations	Historical VaR 95%	Historical VaR 99%	Sharpe Ratio
500	246	143	111	91	70	0.00140174896611285	0.0020242698225955	0.918820176905784

Figure 12 – DAX Studio consolidated query: signal counts, Total ML Predictions 91, Total RL Observations 70, Historical VaR 95% and 99%, and Sharpe Ratio. *Source: project DAX Studio session (E1).*

20.3 Reconciliation Findings

1. **Signal counts.** The counts 500 / 246 / 143 / 111 are identical in the table-level query (Figure 1), the measure queries (Figures 11 and 12), the Python distribution reported in the brief, and the KPI cards on Figures 4 and 7. The donut chart on Figure 4 is the exception (Chapter 18.3).
2. **Historical VaR and Sharpe.** Figures 10 and 12 return identical values for the same measures. The Sharpe Ratio of 0.918820176905784 corresponds to the 0.92 shown on the Risk Management page (Figure 5).
3. **ML and RL counts.** The 91 ML predictions in Figure 2 match Figure 12; the 70 RL observations match the roughly 70 rows visible on Figure 9.
4. **Risk table versus measures.** Historical VaR in the stored Risk_Fact table differs marginally from the Historical VaR measure (Table 16.2); the largest relative difference is about 0.06%.
5. **Display versus value.** Several dashboard cards round to 0.00 while DAX Studio returns non-zero values, demonstrating that the calculations are populated and the defect is purely one of formatting.

The supplied evidence confirms exact agreement between Python and Power BI **only for the signal counts**. Agreement of Python and R outputs, and of Python or R VaR, CVaR, ML and RL values with the Power BI values, is **Not available in the provided project evidence** and is not claimed.

20.4 Limitations

The screenshots show a subset of the queries performed. Measure definitions for risk, ML and RL are not visible. DAX Studio validates internal model consistency; it cannot validate the correctness of the upstream analytical method.

21.0 Gamification Framework

21.1 Objective

To define a transparent, level-based learning and assessment framework that rewards sound analytical and risk behaviour rather than raw profit.

21.2 Methodology

Eight progressive levels map to the project's skills sequence, from basic market literacy to quantitative modelling. A 100-point scoring framework weights signal quality most heavily, followed by risk management, with profitability weighted at 20 points, below the combined 65 points assigned to accuracy and risk management.

Table 21.1 – Gamification levels

Level	Name	Score range	Competency focus (E3, indicative)
1	Forex Beginner	0–20	Market basics and terminology
2	Technical Analyst	21–30	Indicators and chart reading
3	Signal Hunter	31–40	Combining signals
4	Backtest Trader	41–50	Cost-aware backtesting
5	ML Trader	51–65	Machine-learning signals and validation
6	Risk Manager	66–75	VaR, CVaR and position sizing
7	AI Trader	76–90	RL and integrated AI strategies
8	Quant Master	91–100	Full pipeline mastery and validation

Table 21.2 – Scoring framework

Component	Points
Signal Accuracy	40
Risk Management	25
Profitability	20
Drawdown Control	10
Consistency	5
Total	**100**

21.3 Technical Explanation

The total score is the sum of component scores, and the level is a step function of the total:

```
Score = \sum_{j=1}^{5} P_j, \quad P = (P_{acc}, P_{risk}, P_{profit}, P_{dd}, P_{cons}) \le (40, 25, 20, 10, 5), \qquad Level = f(Score)
```

The band widths are unequal (21, 10, 10, 10, 15, 10, 15 and 10 points), so progression is slower in some tiers than in others. The method for converting raw metrics (for example accuracy or drawdown) into component points, and the rounding rule for non-integer scores, are **Not available in the provided project evidence**.

21.4 Results, Interpretation and Limitations

The framework is a design artefact (E2). No player data, leaderboard or user-testing results exist in the supplied evidence, so no engagement or learning-outcome claim is made. A design observation: 65 of 100 points depend on accuracy and risk, but with a highly noisy market, accuracy near chance (as in Chapter 13.4) would cap a participant at the lower levels, which is a property of the scheme worth noting. Profitability should be assessed out-of-sample and after costs to avoid rewarding in-sample over-fitting.

22.0 Business Model (Illustrative)

22.1 Status Statement

This chapter is an "Illustrative Business Model / Projection Framework". It contains no actual company revenue, customers, pricing or market-size data. Everything below is an assumption (class E3) or an empty template to be completed by the author. None of it is an implemented project result.

22.2 Objective

To outline how the analytical prototype could be commercialised as an educational and decision-support product, and to provide a structure for building projections once assumptions are justified.

22.3 Illustrative Business Model Canvas

Element	Illustrative content (assumption)
Value proposition	Transparent, explainable signal analytics with risk context and validated dashboards, positioned for education and research
Customer segments	Learners and analytics students; trading-education providers; analytics teams evaluating BI-based monitoring
Channels	Online courses, institutional partnerships, open-source repository with a documentation site
Revenue streams	Subscription tiers, institutional licences, paid training or certification linked to the gamification levels
Key resources	Codebase, PBIX, validation workbook, documentation
Key costs	Data licensing, cloud and BI licences, development and support time, compliance review
Key risks	Regulatory classification, data licensing restrictions, weak predictive evidence, reputational risk from misuse

22.4 Regulatory and Compliance Considerations

Distributing trading signals may be treated as investment advice or research in some jurisdictions, and retail participation in leveraged FX is restricted or regulated in many countries. Requirements differ by location and must be confirmed with qualified legal counsel before any commercial activity. Positioning the product as education and analytics, with prominent risk disclaimers, reduces but does not eliminate this exposure.

22.5 Projection Framework

Projections should be built bottom-up from stated assumptions. Monthly recurring revenue across tiers k, with N_k subscribers at price P_k, is:

$$MRR_t = \sum_{k} N_{k,t}\,P_k, \qquad N_{k,t+1} = N_{k,t}(1-\chi_k) + A_{k,t}$$

where chi_k is monthly churn and A_k are new acquisitions. Unit economics and break-even are:

$$LTV = \frac{ARPU \times GM}{\chi}, \qquad Payback = \frac{CAC}{ARPU \times GM}, \qquad N_{BE} = \frac{F}{ARPU \times GM}$$

Here ARPU is average revenue per user, GM gross margin, CAC customer-acquisition cost and F fixed monthly costs.

Table 22.1 – Illustrative assumptions (to be completed)

Variable	Definition	Value
P_k	Price per tier	[INSERT ASSUMPTION AND SOURCE]
A_k	New subscribers per month	[INSERT ASSUMPTION AND SOURCE]
chi_k	Monthly churn	[INSERT ASSUMPTION AND SOURCE]
GM	Gross margin	[INSERT ASSUMPTION AND SOURCE]
CAC	Acquisition cost	[INSERT ASSUMPTION AND SOURCE]
F	Fixed monthly costs	[INSERT ASSUMPTION AND SOURCE]

22.6 Limitations

No market research, willingness-to-pay evidence or competitor analysis is provided. Because the ML and RL components have not demonstrated predictive skill in the supplied evidence, any commercial claim about signal quality would be unsupported.

23.0 Excel Validation

23.1 Objective

To provide a transparent, spreadsheet-based cross-check that reviewers without Python, R or Power BI can inspect.

23.2 Methodology and Status

An Excel validation workbook is a mandatory deliverable. The workbook itself, its formulas and its outcomes were **not** part of the supplied evidence; therefore this chapter documents the intended validation design (E3) and records the outcome as **Not available in the provided project evidence**. No claim of agreement with Python, R or Power BI is made on the basis of Excel.

Table 23.1 – Excel validation scope

Check	Standard Excel approach (illustrative)	Outcome
Row count and signal counts	COUNTA and COUNTIF on the Composite_Signal column	Not available in the provided project evidence
Composite score	SUMPRODUCT of component signals and the weight vector (0.15, 0.15, 0.15, 0.20, 0.10, 0.15, 0.10)	Not available in the provided project evidence
Signal class	Nested IF applying the ±0.30 thresholds	Not available in the provided project evidence
Average return and volatility	AVERAGE and STDEV.S on the return column	Not available in the provided project evidence
Sharpe Ratio	AVERAGE / STDEV.S multiplied by the chosen annualisation factor	Not available in the provided project evidence
Historical VaR	PERCENTILE.INC or PERCENTILE.EXC on returns	Not available in the provided project evidence
CVaR	AVERAGEIF of returns at or below the VaR threshold	Not available in the provided project evidence
ML confusion counts	COUNTIFS on actual and predicted columns	Not available in the provided project evidence

23.3 Technical Explanation

Percentile conventions matter. PERCENTILE.INC interpolates over the closed interval [0, 1], whereas PERCENTILE.EXC excludes the end points. On a sample of about 500 returns, the two conventions can differ in the fourth or fifth significant digit at the 99% level. This is one plausible reason for the small Historical VaR differences in Table 16.2, together with

the conventions used in Python (for example linear interpolation) and DAX (PERCENTILE.INC or PERCENTILE.EXC). The convention used in each tool is **Not available in the provided project evidence**.

23.4 Limitations

A spreadsheet check is only as independent as its inputs: if it reads the same processed CSV, it validates the calculations but not the upstream data preparation. The workbook should record the percentile convention, annualisation factor and the file version it reads.

24.0 End-to-End Validation

24.1 Objective

To assess consistency across Python, R, Power BI, DAX Studio and Excel for data, signals, risk metrics, model outputs and dashboards, and to document discrepancies openly.

24.2 Methodology

Each validation dimension was evaluated against the available evidence. A cell is marked “Confirmed” only where a screenshot or the brief supports it; otherwise it is marked “Reported” (author-stated, no exhibit) or “Not available”.

Table 24.1 – Validation matrix

Dimension	Python	R	Power BI	DAX Studio	Excel
Row count (500)	Reported	Not available	Confirmed (Figures 4, 7)	Confirmed (Figures 1, 10–12)	Not available
Signal counts 246 / 143 / 111	Reported	Validation reported; values not available	Confirmed (KPI cards; donut incorrect)	Confirmed	Not available
Composite weights and thresholds	Reported	Reported	Not visible	Not visible	Not available
Return, volatility, Sharpe	Not available	Not available	Displayed (0.92; rounding issues)	Confirmed values (Figure 10)	Not available
Historical VaR	Not available	Computed (values not available)	Displayed as 0.00 on cards	Confirmed; table differs slightly	Not available

Dimension	Python	R	Power BI	DAX Studio	Excel
Monte Carlo VaR, CVaR	Not available	Computed (values not available)	Displayed as 0.00 on cards	Confirmed (Figure 3)	Not available
ML outputs	Reported	Reported (caret, XGBoost)	Not applicable	Confirmed counts (Figure 2)	Not available
RL outputs	Reported	Not applicable	Observations visible (Figure 9)	Confirmed count 70 (Figure 12)	Not available

24.3 Findings by Dimension

Data consistency. The 500-row sample is consistently reflected across Power BI and DAX Studio. Weekend rows and an unlabelled periodicity (hourly rows versus the “daily” labels on the measures) are unresolved data-definition issues.

Signal consistency. The composite signal distribution agrees exactly between Python (as reported), the Power BI table and the DAX measures. Agreement of the R reproduction is reported as validated, with figures unavailable.

Risk-metric reconciliation. DAX Studio confirms a full set of VaR, CVaR and Kelly values. Within the model, the measure and the stored table differ by at most about 0.06% for Historical VaR. Cross-tool reconciliation with Python, R and Excel is not documented.

Model-output comparison. The ML model made 91 predictions with near-chance accuracy and almost no positive predictions; the PPO policy was long in about 87% of the roughly 70 comparison rows; the technical signal was split between BUY and SELL. The three approaches therefore disagree materially, and none has demonstrated profitability in the evidence.

Dashboard validation. Values in KPI cards were checked against DAX Studio. Calculation values are correct where compared; the principal problems are formatting and chart configuration.

24.4 Known Discrepancies and Observations

Table 24.2 – Known discrepancies and observations

ID	Observation	Evidence	Severity	Recommended action
D1	Signal donut shows 500 (33.33%) for BUY, HOLD and SELL	Figure 4	High	Rebuild the chart using a row count by Composite_Signal

ID	Observation	Evidence	Severity	Recommended action
D2	ML predicts almost exclusively the negative class (TP 1, FN 42); ML signal constant at -1	Figures 2, 9	High	Investigate target definition, class balance and threshold; report honestly
D3	RL evaluation did not mark open positions to market or force-close final positions	Project brief	High	Re-run evaluation with terminal liquidation and costs
D4	Annualisation and “daily” labelling versus hourly rows	Chapter 16.5	Medium	Confirm periodicity; state the annualisation factor in the measure
D5	VaR, CVaR, volatility and range cards display 0.00	Figures 5, 8	Medium	Use percent format with three decimals
D6	Weekend observations present	Figure 7	Medium	Confirm data-vendor convention; consider excluding
D7	Historical VaR measure differs slightly from Risk_Fact	Table 16.2	Low	Align the percentile convention and input series
D8	Metric table total of 0.12 adds unlike metrics	Figure 8	Low	Remove the total
D9	Kelly fractions share an axis with VaR values	Figure 8	Low	Separate the visuals
D10	Title spellings “Seasson” and “Comparision”	Figures 7, 8, 9	Low	Correct the labels
D11	Redundant volume charts; uninformative average-close-by-session bars; BUY KPI shown as bar	Figures 4, 6, 7	Low	Simplify visuals

ID	Observation	Evidence	Severity	Recommended action
D12	Pair Analysis averages differ from all-row averages (1.35 versus 1.4056 for spread)	Figures 6, 11	Information	Add a visible filter indicator to the page

24.5 Limitations

Five-tool agreement could not be established because Python, R and Excel numeric outputs were not part of the evidence. The validation is therefore partial, and only the claims marked “Confirmed” are supported by exhibits.

25.0 Results and Discussion

25.1 Summary of Evidenced Results

Table 25.1 – Summary of evidenced results

Result	Value	Source	Note
Observations	500	Figures 1, 10–12	EUR/USD only
BUY / SELL / HOLD	246 / 143 / 111	Figures 1, 11, 12	49.2% / 28.6% / 22.2%
Average Close	1.088007	Figure 11	All rows
Average Spread	1.4056	Figure 11	Unit not stated
Average Daily Return	5.242×10^{-5}	Figure 10	Periodicity to be confirmed
Sharpe Ratio (dashboard)	0.9188	Figure 10	Basis undocumented
Historical VaR 95% / 99% (measure)	0.001402 / 0.002024	Figure 10	Per observation period
Parametric VaR 95% / 99%	0.001439 / 0.002057	Figure 3	

Result	Value	Source	Note
Monte Carlo VaR 95% / 99%	0.001448 / 0.002051	Figure 3	
CVaR 95% / 99%	0.001798 / 0.002431	Figure 3	
Kelly / Half-Kelly	0.07 / 0.03	Figure 8	Display precision
ML predictions	91 (48 correct)	Figure 2	Accuracy 52.7% (derived)
RL observations	70	Figure 12	PPO net signal 53
Session counts	Asia 168; London 105; Overlap 105; New York 82; Off 40	Figure 7	

25.2 Discussion

The project succeeded as an **engineering and analytics-integration exercise**: a single set of signals flows from Python through Power BI, is queryable and reconcilable in DAX Studio, and is presented across six pages with a risk framework attached. The signal counts reconcile exactly, and the risk metrics are internally coherent (CVaR above VaR, 99% above 95%, and three VaR methods within a few percent of each other).

The project did **not** demonstrate predictive or economic value. The ML model's accuracy equals that of an all-negative classifier. The RL evaluation is methodologically incomplete. Backtest metrics are not available in the evidence. The composite signal's BUY dominance coincides with an upward drift in the sample and cannot be separated from simple trend exposure.

Objective status: objectives 1–2 (EDA, indicators and composite signal) are evidenced in part; objective 3 (backtesting) is reported but unevidenced; objectives 4–5 (ML, RL) are implemented with weak or incomplete evidence; objective 6 (case studies) is qualitative; objective 7 (risk) is evidenced; objective 8 (R) is reported; objectives 9–10 (Power BI, validation) are evidenced with identified defects; objective 11 (gamification, business model) is a framework only; objective 12 (report and handover) is addressed in Chapter 30.0.

25.3 Implications

For a practitioner, the principal value of the prototype lies in its reproducible pipeline and explicit risk reporting. For a researcher, the key next step is to repeat the analysis on a longer, multi-pair, multi-regime dataset with correct out-of-sample evaluation. Nothing in the results justifies live trading.

26.0 Limitations

26.1 Overview

The limitations below are consolidated here and cross-referenced to the chapters where they arise. The dataset limitation is the most consequential and conditions every other result.

Table 26.1 – Limitation register

ID	Limitation	Consequence	Reference
L1	Dataset contains only EUR/USD; the project did not use five pairs	No cross-pair correlation, portfolio risk or generalisation	Ch. 8.0, 9.0, 16.0
L2	Approximately 500 hourly rows (about three weeks); the project did not use two years of data	Single regime; unstable statistics; weak validation	Ch. 8.0
L3	Weekend observations and unlabelled periodicity (“daily” labels on hourly rows)	Annualised metrics may be mis-scaled	Ch. 9.5, 16.5
L4	Backtest metrics not available in the evidence	Strategy merit cannot be assessed	Ch. 12.4
L5	ML accuracy equals the all-negative baseline; recall 2.3%	No demonstrated predictive skill	Ch. 13.5
L6	RL evaluation did not fully mark open positions to market and did not always force-close final positions	RL profitability must not be inferred	Ch. 14.5
L7	Fixed, unvalidated composite weights and thresholds	Regime sensitivity; no calibration	Ch. 11.6

ID	Limitation	Consequence	Reference
L8	Parametric VaR assumes normality; historical VaR is sample-dependent; Monte Carlo single-pair	Understates tail and regime-break risk	Ch. 16.0
L9	Case-study P&L not available	Case studies are qualitative	Ch. 15.0
L10	Dashboard defects (donut chart, rounding, redundant visuals)	Risk of misreading the dashboard	Ch. 18.0, 24.4
L11	Python/R/Excel numerical agreement not documented	Partial validation only	Ch. 17.5, 23.0, 24.0
L12	Business model is illustrative	No commercial claims	Ch. 22.0
L13	Backtest and simulation outcomes do not guarantee future performance	Not a basis for trading decisions	Throughout

27.0 Ethical and Financial Risk Considerations

27.1 Not Financial Advice

This report and the associated prototype are analytical and educational. They do not constitute investment, legal or tax advice, and they are not a recommendation to buy or sell any instrument. The system is not a guaranteed profitable trading system.

27.2 Financial Risk

Leveraged FX trading can produce losses exceeding the initial deposit. The 2015 EUR/CHF event (Chapter 15.6) illustrates that gaps can overwhelm stop-losses and that losses can propagate to intermediaries. VaR and Kelly estimates computed from three weeks of data are unreliable guides to position size.

27.3 Model and Evidence Integrity

Overfitting, data snooping and selective reporting are major risks in quantitative finance. This report therefore separates evidence classes, discloses null or negative findings (ML near-chance accuracy, incomplete RL evaluation) and avoids unsupported numbers. Any

extension should use pre-registered evaluation protocols, strict out-of-sample testing and multiple-testing adjustments.

27.4 Responsible Use of Gamification

Gamified scoring can encourage excessive risk-taking or addictive engagement. The framework's emphasis on risk management and drawdown control, and its use of simulated rather than live capital, should be preserved. Any deployment should include risk warnings, cooling-off features and age and suitability controls.

27.5 Data, Privacy and Licensing

The supplied evidence contains no personal data. Market-data licences may restrict redistribution or commercial use; the source licence for the EUR/USD data is **Not available in the provided project evidence** and should be confirmed before publication or commercialisation.

27.6 Transparency and Accountability

The composite engine is rule-based and inspectable, which supports explainability, whereas the LSTM and PPO components are harder to interpret. SHAP mitigates this partially for tree models. Dashboards should state their data window, sample size and known defects so that viewers are not misled by precise-looking numbers on a small sample.

28.0 Future Enhancements

Data and scope

- Acquire multi-year, multi-pair data (for example five major pairs) and re-run every stage; this was not achieved in the present project.
- Resolve periodicity and weekend-bar handling; document the data vendor and licence.

Methodology

- Compute and report backtest metrics with confidence intervals, a deflated Sharpe Ratio and bootstrap tests, and compare against buy-and-hold and random-signal baselines.
- Estimate or optimise composite weights and thresholds within a nested walk-forward scheme; run sensitivity analyses.

- Add regime detection (trend versus range, volatility regimes) and volatility-scaled position sizing.
- ML: address class imbalance, calibrate probabilities, tune decision thresholds, and report per-model metrics and SHAP rankings.
- RL: fix the evaluation (mark-to-market, forced terminal liquidation), use multiple seeds and report distributions of outcomes.
- Risk: correlated multi-pair Monte Carlo using a covariance matrix, stress scenarios inspired by Chapter 15.0, and backtesting of VaR (Kupiec and traffic-light tests).

Engineering and BI

- Fix the dashboard defects listed in Table 24.2; format risk cards as percentages; add a measure dictionary.
- Automate the pipeline with tests, version-pinned environments and continuous integration.
- Add paper-trading (forward) validation before any live use, with regulatory review.

29.0 Conclusion

The project delivered an end-to-end analytical prototype for EUR/USD that integrates data preparation, a weighted composite signal, cost-aware backtesting, machine learning, reinforcement learning, risk metrics, an R reproduction and a six-page Power BI report, together with a DAX Studio reconciliation. The strongest evidenced outcome is internal consistency: 500 observations and 246 BUY, 143 SELL and 111 HOLD signals reconcile across the Python output, the Power BI table and DAX measures, and a coherent set of VaR, CVaR and Kelly metrics is stored and displayed.

The evidence does not show predictive skill or profitability. The ML output is indistinguishable from an all-negative classifier, the RL evaluation has a known methodological gap, backtest metrics and case-study P&L are not available, and the dataset is far narrower than originally envisaged (one pair and about 500 rows, rather than five pairs and two years). The honest reading is that the project is a well-structured foundation and a documented learning exercise whose findings should be treated as exploratory. The defects identified in Chapter 24.0 are tractable, and Chapter 28.0 provides a path to a more rigorous evaluation. Past or simulated performance does not guarantee future results.

30.0 Handover and Deployment

30.1 Objective

To enable another analyst to reproduce, audit and extend the project without the original author. The repository structure below is a **recommended** layout; the actual repository contents were not supplied, and only forex_eda_processed.csv and forex_signals.csv are named in the project evidence.

30.2 Deliverable Inventory

The handover package comprises the GitHub repository; the Python codebase; the R codebase; the Power BI PBIX file; the Excel validation workbook; this final report; the presentation; and project documentation (README, data dictionary, measure dictionary, validation log).

30.3 Recommended Repository Structure

```
forex-signal-ai-agent/
  README.md
  LICENSE
  data/
    raw/                        (source OHLCV file; name not available)
    processed/forex_eda_processed.csv
    signals/forex_signals.csv
  python/
    day02_eda/
    day03_signals/
    day04_backtest/
    day05_ml/
    day06_rl/
    day08_risk/
  r/
    day09_r_implementation/   (tidyverse, caret, xgboost, Shiny app)
  powerbi/
    ForexSignalAgent.pbix     (proposed name)
    measure_dictionary.md
  validation/
    dax_studio_queries/
    excel_validation.xlsx     (proposed name)
    validation_log.md
  docs/
```

```
    final_report/
    presentation/
    case_studies/
  requirements.txt / renv.lock (environment pinning)
```

30.4 Knowledge-Transfer Checklist

Table 30.1 – Knowledge-transfer checklist

#	Item	Owner	Status
1	Repository access and branch conventions explained	[INSERT]	Open
2	Python environment build and run instructions verified on a clean machine	[INSERT]	Open
3	R environment (renv or equivalent) restored and Shiny app launched	[INSERT]	Open
4	Data provenance, licence, periodicity and weekend-bar handling documented	[INSERT]	Open
5	Component signal rules, weights and thresholds documented	[INSERT]	Open
6	Backtest parameters, costs and metrics recorded	[INSERT]	Open
7	ML target definition, hyper-parameters and metrics recorded	[INSERT]	Open
8	RL reward, window length, seeds and corrected evaluation recorded	[INSERT]	Open
9	Power BI model view, relationships and measure dictionary (120 measures) delivered	[INSERT]	Open
10	Dashboard defects D1–D12 triaged	[INSERT]	Open
11	DAX Studio queries saved and reproducible	[INSERT]	Open
12	Excel workbook formulas, percentile convention and annualisation factor documented	[INSERT]	Open
13	Cross-tool reconciliation table completed (Python, R, Power BI, Excel)	[INSERT]	Open

#	Item	Owner	Status
14	Risk disclaimer and “not financial advice” wording confirmed in all deliverables	[INSERT]	Open

30.5 Final Demonstration Plan

A suggested twenty-minute session:

1. Context, scope and prominent limitations (2 minutes).
2. Data description and EDA outcomes, including weekend rows (2 minutes).
3. Signal engine: weights, thresholds and the Figure 1 table-level query (3 minutes).
4. Backtesting, ML and RL: what exists and what the evidence does and does not show (4 minutes).
5. Power BI walk-through of the six pages, including the identified defects (4 minutes).
6. DAX Studio reconciliation: Figures 10–12 and the Historical VaR difference (3 minutes).
7. Roadmap and questions (2 minutes).

30.6 Deployment Notes

The prototype is batch-oriented and not deployed. A pilot deployment would require scheduled data refresh, secured Power BI workspace publishing, access control, monitoring of data quality, and a legal review (Chapter 22.4). Run commands, scheduler settings and hosting details are **Not available in the provided project evidence**.

31.0 References

The references below are standard methodological sources. Citation details should be verified against the originals before submission, and the author should add primary sources (central-bank statements, data-vendor documentation and university formatting requirements) that were not part of the supplied project evidence.

1. Appel, G. (2005). *Technical Analysis: Power Tools for Active Investors*. FT Press.
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32.0 Appendices

Appendix 1 – Evidence Index (screenshots supplied, in upload order)

Screenshot	Content	Used as	Chapter
1	DAX Studio: measure catalogue query; Total Signals, Average Daily Return, Sharpe Ratio, Historical VaR	Figure 10	20.0
2	Power BI Pair Analysis page	Figure 6	18.5
3	DAX Studio: ML prediction measures	Figure 2	13.4
4	Power BI Strategy Comparison page	Figure 9	18.8
5	DAX Studio: consolidated signal, ML, RL and risk measures	Figure 12	20.0
6	Power BI Risk Management page	Figure 5	18.4
7	Power BI Signal Overview page	Figure 4	18.3
8	Power BI Session Heatmap page	Figure 7	18.6
9	DAX Studio: signal counts, Average Close, Average Spread	Figure 11	20.0
10	DAX Studio: COUNTROWS / CALCULATE signal-count query	Figure 1	11.4
11	Power BI Monte Carlo & VaR page	Figure 8	18.7
12	DAX Studio: Risk_Fact VaR and CVaR rows	Figure 3	16.4

Appendix 2 – Placeholder Figure Register

Placeholder	Intended content	Chapter
P-1	End-to-end system architecture diagram	6.0
P-2	Power BI model view with relationships	7.0

Placeholder	Intended content	Chapter
P-3	Return distribution, return series and rolling volatility	9.0
P-4	Price with moving averages, Bollinger Bands and Ichimoku overlays	10.0
P-5	Equity curve and drawdown: MA-only versus composite strategy	12.0
P-6	SHAP summary plot and model comparison	13.0
P-7	PPO and DQN training curves and held-out evaluation	14.0
P-8	R Shiny dashboard prototype	17.0
P-9	Excel validation workbook overview	23.0

Appendix 3 – Data Dictionary (fields evidenced in the project material)

Field	Description	Source
DateTime	Timestamp of the observation	Brief; Figures 6, 9
Open, High, Low, Close	Price components	Brief; Figure 6
Volume	Traded volume (unit not stated)	Brief; Figure 6
Spread	Bid–ask spread (unit not stated)	Brief; Figure 11
Composite_Signal	BUY, SELL or HOLD class	Figure 1
Metric, Value	Key–value rows in Risk_Fact	Figure 3
PPO Signal, ML Signal, Technical Signal, Next Return	Strategy-comparison fields (+1, 0, -1; return)	Figure 9

Appendix 4 – Formula Summary

See Chapters 9.3, 10.3, 11.3, 12.3, 13.3, 14.3, 16.3, 21.3 and 22.5 for the formulas used in this report.

Annex A – Final Deliverables Checklist

Table A.1 – Deliverables checklist

#	Mandatory deliverable	Evidence in the supplied material	Status
1	30+ page final report	This document (draft for author review; confirm page count after formatting)	Drafted
2	Python codebase	Reported; not supplied	To be confirmed
3	R codebase	Reported; not supplied	To be confirmed
4	Power BI dashboard	Six-page screenshots supplied; PBIX not supplied	Partly evidenced
5	120+ DAX measures	Author-reported count; partial names visible	To be confirmed
6	DAX Studio validation	Screenshots supplied (Figures 1–3, 10–12)	Evidenced
7	Excel validation workbook	Not supplied	To be confirmed
8	Final presentation	Not supplied	To be confirmed
9	GitHub / project handover documentation	Recommended structure and checklist in Chapter 30.0	Drafted

Annex B – Dataset Limitations

- Only EUR/USD is available; the project did not use five currency pairs.

- Approximately 500 hourly rows covering about three weeks; the project did not use two years of history.
- Single short regime; no rate-decision, crisis or low-liquidity episodes of significance.
- Weekend observations present; periodicity labelling inconsistent with hourly rows.
- Data vendor, licence, spread unit and volume unit are not documented.
- Rolling cross-pair correlation and multi-pair correlated Monte Carlo analysis could not be performed.

Annex C – Known Technical Limitations

- ML output is near-chance and almost exclusively negative-class (TP 1, FN 42 of 91).
- RL evaluation did not fully mark open positions to market and did not always force-close final positions; profitability is not established.
- Backtest metrics, ML per-model metrics, case-study P&L and Python/R/Excel numerical comparisons are not available in the provided project evidence.
- Dashboard defects: incorrect donut chart, VaR and CVaR cards rounding to 0.00, an unmeaningful metric total, redundant or uninformative visuals and label typos (Table 24.2).
- Sharpe Ratio and annualised volatility use an annualisation basis that is not documented and may not match the hourly periodicity.
- Historical VaR differs marginally between the DAX measure and the stored table (about 0.04% to 0.06%).

Annex D – Future Work

1. Rebuild the dataset with multiple pairs and multiple years, then repeat every stage.
2. Correct and expand the evaluation (backtest with confidence intervals, RL with terminal liquidation, ML per-model metrics, baselines).
3. Add regime-aware weighting, volatility-scaled sizing and stress scenarios based on the four case studies.
4. Implement correlated multi-pair Monte Carlo VaR and VaR backtesting.
5. Fix the dashboard defects and publish a measure dictionary.
6. Automate the pipeline and complete cross-tool reconciliation including Excel and R.

Annex E – Appendix: File and Folder Structure

The recommended repository tree is given in Chapter 30.3. Artefacts named in the project evidence are:

```
forex_eda_processed.csv     Processed EDA dataset (Day 2)
forex_signals.csv           Indicator values, component signals and composite
signal (Day 3)
Power BI report (PBIX)      Eight tables, 120 DAX measures, six pages (Days
10–11)
DAX Studio queries          Validation and reconciliation queries (Day 12)
Excel validation workbook   Spreadsheet cross-checks (Day 14)
Python codebase             EDA, signals, backtesting, ML, RL, risk (Days 2–
8)
R codebase                  tidyverse, caret, xgboost, Monte Carlo, Shiny
(Day 9)
```

End of report.